Erisma — User manual

Version 0.4.0

Contents

1. What Erisma is
2. Installation and start
3. The screen
4. Entering data
5. Saving, files and printing
6. National Annex and project site
7. Reading the report
8. The 3D model
9. Verification of Erisma Structure
10. Erisma Structure: whole buildings
11. Projects: several calculations together
12. Linking calculations (calculation chain)
13. Project report and company logo
14. Quantities and cost estimate
15. Automatic design — most economical solution
16. Comparing alternative solutions
17. DXF drawings for CAD
18. Trial and licence
19. Keyboard shortcuts
20. National Annexes in Erisma
21. Calculation reference

- Continuous beam analysis
- RC building — 3D analysis and design
- Pile cap (strut-and-tie)
- RC beam section design
- RC column
- RC shear wall (EC8 DCM)
- RC slab
- RC stair flight
- Water tank / pool
- Assessment and strengthening of RC members
- Fire resistance — RC beam / slab
- Fire resistance — steel member
- Fire resistance — timber beam
- Berlin wall (soldier pile wall)
- Cantilever retaining wall
- Embedded pile wall with anchors
- Foundation settlement
- Gabion wall
- Gravity retaining wall
- Liquefaction potential
- Pad footing
- Raft foundation on springs
- Single pile — axial and lateral
- Slope stability (Bishop)
- Strip footing and tie beams
- Floor and roof loads
- Load combinations
- Wind and snow loads at the site
- Masonry wall
- Bolted connection
- Column base plate
- Fillet weld group
- Steel beam design
- Steel column
- Timber beam

1. What Erisma is

Erisma is a suite of structural and geotechnical calculations to the Eurocodes. Every calculation is written out step by step — **input → formula → value → clause → check** — so that the engineer can verify each number by hand and submit the calculation sheet.

- **Results while you type:** checks, report and 3D model update after every change.
- **Worked examples as defaults:** every calculation opens with realistic values that pass, so you always start from a sensible design and change only what differs.

- **Hints:** every failing check says what to change (“increase the base width…”).
- **National Annexes:** partial factors and nationally determined parameters are read from the annex chosen in the project details and listed in every report.
- **Verified:** each module is tested against hand calculations and published values.

> Erisma is a tool for qualified engineers. The engineer remains responsible for the model, the input values and the interpretation of the results.

2. Installation and start

Windows — installer (recommended)

1. Run `Erisma-Setup-version.exe`. Python or anything else is not needed. Choose the language (Greek / English) and, if you wish, a desktop icon. No administrator rights are required (it installs for the current user; an administrator can choose all users).
2. Start **Erisma** from the Start menu or the desktop. A small window “Erisma — στατικοί υπολογισμοί” appears and the program opens in your browser. The window shows the address (normally `http://127.0.0.1:8000`; if that port is busy the next free one is used). **Άνοιγμα** opens the program again in the browser, **Τερματισμός** or closing the window stops Erisma.
3. The Start menu also has **Εγχειρίδιο Erisma** (this manual as PDF) and the uninstaller. Updating = running the new Setup; your licence and settings are kept.

Without installer (Windows, Mac, Linux)

1. Install Python 3.10 or newer (on Windows tick *Add Python to PATH*).
2. Unzip `Erisma.zip` into a folder of your choice.
3. **Windows:** double-click `Erisma.bat`. **Mac:** right-click `Erisma.command` → Open.

The first start installs the program (1–2 minutes, internet needed once); then Erisma opens in your browser at `http://127.0.0.1:8000`. To stop it close the black window.

Either way everything runs on your own computer and works offline; nothing is sent to the internet. Your projects are ordinary `.erisma` files that you keep wherever you like.

3. The screen

- **Home page:** all calculations as cards grouped by category, with search. “Continue” reopens the last calculation.
- **Left sidebar:** search (`Ctrl`+`K`), ★ favourites, recent calculations and all categories. « collapses it to icons.
- **Project details** (top of the form): company, project, job no., element, engineer, checker, date, **National Annex** and **project site**. They appear in the title block of every report and are kept for all calculations.
- **Input form** (middle): fields grouped in boxes. Above it the **status banner**: green when all checks pass, red with the failing checks and what to change.

- **Report / 3D model** (right): the full calculation sheet and the interactive model.
- **Above the form: + Add to project** (see "Projects"), ⚖ **Alternative** (see "Comparing alternative solutions"), 🎯 **Most economical** (automatic design), 📖 **How it works** (this manual at that calculation) and ↺ **Reset** (back to the example values). In the status banner the 💶 cost estimate.
- **Top bar:** Home, light/dark theme, language ΕΛ/EN, licence state, Manual, Open, Save, **DXF** (only for calculations with a drawing), Export HTML, Print / PDF.
- At the top of the left sidebar the 📁 **Project** list with the buttons 📄 project report, ⚖ comparison and ✚ new project.

4. Entering data

- **Decimals:** type `1,5` or `1.5` — both work.
- **Arithmetic:** type `2*3,5`, `(4+2)/2` or `1,5^2`; the result is shown next to the field and replaces the expression when you leave it.
- **Units** are shown next to every field; type the value in that unit.
- **Keyboard:** `Enter` goes to the next field, `Shift`+`Enter` back; `↑`/`↓` change the value in small steps (with `Shift` ×10).
- **Help:** the help text of a field appears while the field is selected, or click ⓘ.
- **Changed fields** are marked with a dot; ↺ returns that field to the example value. Each group shows how many of its fields you changed. "↺ Reset" restores the whole example.
- **Groups** can be collapsed by clicking their title — Erisma remembers it.
- **Conditional fields** appear only when needed (e.g. jacket data only when "RC jacket" is chosen).
- **Tables** (soil layers, anchors, loads…): "+ Add row" copies the last row, ⧉ duplicates a row, × deletes it. You can **paste a block of cells from Excel**: select the first cell and press `Ctrl`+`V`.
- **Undo / redo:** `Ctrl`+`Z` / `Ctrl`+`Y`.
- **Wrong values** turn red immediately; if the calculation cannot run, the banner says why and the field concerned is highlighted.

5. Saving, files and printing

- **Autosave:** the inputs of every calculation are kept in the browser; when you reopen a calculation you continue where you left off.
- **Save** (`Ctrl`+`S`) downloads a file `.erisma` with the inputs and the project details — the whole project (all items, links, logo, unit prices) when the project list is not empty, otherwise the current calculation. **Open** (`Ctrl`+`O`) loads either kind again — send it to a colleague or keep it with the job.
- **Print / PDF** (`Ctrl`+`P`) prints the report; choose "Save as PDF" as the printer to get a PDF. **Export HTML** saves the report as a web page.

6. National Annex and project site

Choose the **National Annex** in the project details. All partial factors and nationally determined parameters (NDPs) come from it, and each report ends with a table of the NDPs used. Where Erisma has no annex for a part (for example EN 1995-1-1 for Cyprus) the EN recommended values are used and the report says so.

With the Cyprus annex, 📍 **Map** opens the official seismic zonation map: click on the site, search a town, use your location or type coordinates. The seismic zone and a_{gR} are filled in automatically (near a boundary the higher zone within 1 km is used), and the wind and snow calculation reads $v_{b,0}$ and s_k for the same site. Next to every a_{gR} field the zone buttons set the value with one click.

7. Reading the report

- The **title block** carries the project details; the summary lists every check with its utilisation.
- Each line shows **symbol = expression = value unit** with the clause of the standard.
- Checks show demand ≤ capacity, the utilisation U and **PASS / FAIL**; failing checks carry a hint in the status banner.
- Figures (sections, pressure diagrams, spectra) are drawn to scale from the actual values.
- The last table lists the National Annex parameters used.

8. The 3D model

Every calculation has an interactive 3D model (tab “3D model” or `Ctrl`+`3`). Drag to rotate, right-drag to pan, scroll to zoom. The **Layers** panel switches soil, loads, pressures, reinforcement, deflected shapes and labels on and off; the buttons give 3D, front and plan views and **PNG** saves an image for your report. The panel on the right lists the utilisation of every check; the model updates with every change.

9. Verification of Erisma Structure

As the verification manuals of commercial programs, this table compares Erisma with independent reference values — closed-form solutions and tabulated results from the literature. The values are computed now, by the program you are using; the same problems run in the automatic test suite before every release.

To compare with your own projects, model the same building in Erisma Structure and in your current program and compare the periods, base shears, member forces and reinforcement — and please send us any difference you find.

No.	Problem	Reference (source)	Reference	Erisma	Difference

1	Cantilever, tip load — tip deflection	$PL^3/3EI$ (Timoshenko & Gere)	5.357 mm	5.357 mm	+0.00 %
2	Fixed beam, triangular load — end moment	$wL^2/20$ (Roark, Table 8.2)	21.6 kNm	21.6 kNm	-0.00 %
3	Portal with rigid beam — sway	$Hh^3/24EI$	0.2143 mm	0.2143 mm	+0.00 %
4	Two-storey shear building — T_1	$\omega^2 = (3-\sqrt{5})/2 \cdot k/m$ (Chopra §10)	0.2811 s	0.2811 s	+0.00 %
5	Cantilever column — elastic critical load factor	$\pi^2EI/4L^2$ (Euler)	16.58 —	16.58 —	+0.00 %
6	Column with P and H — second-order sway (P-Δ)	$H/(Pk)(\tan kL - kL)$	7.493 mm	7.493 mm	-0.00 %
7	Simply supported square plate, UDL — centre deflection	$0.00406\ qa^4/D$ (Timoshenko & Woinowsky-Krieger, Table 8)	2.526 mm	2.524 mm	-0.05 %
8	Clamped square plate, UDL — centre deflection	$0.00126\ qa^4/D$ (ibid., Table 35)	0.7838 mm	0.7849 mm	+0.13 %
9	Plate on Winkler springs, UDL — settlement	q / k_s	1.25 mm	1.25 mm	-0.00 %
10	EN 1998-1 design spectrum, plateau	a_g S 2.5 / q (Eq. 3.14)	0.1371 g	0.1371 g	+0.00 %
11	Strip footing bearing resistance, φ' = 30°	EN 1997-1 Annex D.4	761.1 kPa	761.1 kPa	+0.00 %

10. Erisma Structure: whole buildings

Erisma Structure is the second space of the program (button on the home page). It analyses and designs a whole RC building drawn in plan, with the same calculations, reports, 3D viewer and licence as the member calculations.

1. **Grid:** on the left give the spans along X and Y and the storey heights. The defaults are a realistic 3-storey building that passes.
2. **Draw:** pick a tool and click in the plan — **Column** (C) or **Wall X / Y** (W) on a grid point, **Beam** (B) between two grid points (adds or removes it), **Opening** (O) on a slab panel. With **Select** (V) click any element to change its size; **Del** removes it, **Ctrl+Z / Ctrl+Y** undo and redo.
3. **Results:** after every change the building is re-analysed (3D frame, modal response spectrum, drift, θ, capacity design) and every member is designed. In the plan each member is coloured by its utilisation; red means it fails — the panel on the right says why and what to change. The table of the selected element lists its reinforcement per storey; ↗ opens that member, with its design forces, in the member calculation.
4. **Report and 3D:** the **3D** and **Report** buttons (Ctrl+3) show the model and the full step-by-step report, which prints with Ctrl+P.

The ribbon follows the workflow of professional structural software — File → Home → Draw → Define → Foundation → Loads → Analyse → Design → Results → View → Help. In the 3D view the member under

the pointer is highlighted with its name, size, bars and utilisation, a click selects it; in Results the chosen mode (or the deformed shape) can be animated, with a speed control. Previously it grouped the commands as: **File** (new, open, save, print, HTML report, CSV table) · **Home** (undo/redo, delete, select all columns/beams/walls or the failing members, **Run** F5 and the **Auto** switch for large models, views) · **Model** (element tools, grid wizard, add/remove axes and storeys, default sections) · **Loads** · **Seismic** · **Design** (materials, options, member table) · **Results** (colour by utilisation, steel ratio or pass/fail, the storey shown, reinforcement labels, jump to modes, storey checks and quantities in the report) · **View** (zoom, fit, what is shown, full screen) · **Help**. Double-click a tab to collapse the ribbon.

In the plan: **Shift+click** adds to the selection, dragging a window selects several elements (their sizes can then be changed together), the wheel zooms, the middle button pans and the right button opens a menu (properties, select all of the type, open in the member calculation, delete). The status bar shows the cursor coordinates and how many members pass.

What is checked. Columns (EC2 bending, biaxial, slenderness, shear; EC8 DCM detailing), beams (EC2 bending, shear, **deflection by span/depth 7.4.2**; EC8 detailing), walls (EC8 ductile walls), **slabs** (every panel as a two-way slab, continuous over neighbouring panels), the **foundation** — a **raft** (default): MITC4 plate finite elements on Winkler springs analysed together with the frame, Wood–Armer bending with a basic mesh and additional bars, shear, punching at every column (EN 1992-1-1 6.4, soil reaction deducted), EC7 bearing and Annex H settlement; or foundation beams on springs; or isolated pad footings; seismic actions with the overstrength $\gamma_{Rd}\cdot\Omega$ of EN 1998-1 4.4.2.6 —, storey drifts, θ, α_{cr}, effective modal mass and capacity design ($\Sigma M_{Rc} \geq 1.3\ \Sigma M_{Rb}$, capacity shears).

Loads and combinations. In the **Loads** tab create load cases (permanent, imposed with its use category, snow), pick the Area / Line / Point tool and click slab panels, beams or grid points (on all floors, the roof or one storey). **Combinations…** lists the EN 1990 combinations generated with the factors of the National Annex (ULS 6.10 with each variable action leading, SLS characteristic / frequent / quasi-permanent, seismic) and lets you add your own; members are designed for the envelope.

Plan + 3D shows both windows at once; elements can be selected in the 3D view too (empty positions are drawn faintly so that the tools can add columns and beams there). The **Results** tab switches the 3D bending-moment, shear, axial-force and deformed-shape diagrams and the first three mode shapes on and off; the right panel lists the periods with the character of each mode (X, Y or torsion) and the EC8 regularity, the left panel the storey drifts. The report contains the full regularity check (centres of mass and stiffness, torsional radii, compactness, slenderness, storey stiffness) and the resulting q.

Auto-size (Home or Design tab) repeats the complete calculation and enlarges what fails — columns, beams, walls, slab thickness, foundation beams, the number of modes, P-Δ — until every Eurocode check passes. It lists every change and applies it only when you press **Apply** (Ctrl+Z undoes it).

The layout is the same on every storey (typical floor). The work is saved automatically in the browser; **Save** writes a project file that **Open** brings back.

11. Projects: several calculations together

A **project** collects all the calculations of one job (e.g. column C1, footing F1, retaining wall W1) so that they are saved, linked, priced and printed together.

1. Open a calculation and set its inputs.
2. Press **+ Add to project** (above the form) and give the element a name, e.g. “Footing F1”. The calculation is copied into the project; from now on the inputs you change are kept in that project item.
3. Repeat for every element. The same calculation can be added several times (F1, F2, …).

The 📁 **Project** section at the top of the left sidebar lists the items. The dot in front of each name is green when all checks pass, red when a check fails and grey before the first run; hover over it to see the maximum utilisation U. 🔗*n* shows how many inputs are linked. Next to each item: ↑ moves it up (this is also the order in the project report), ✎ renames, × removes it (links that used it are removed too). The buttons under the title are 📄 **Report** (project report), ⚖ **Compare** and ✚ **New** (new, empty project).

The project is autosaved in the browser. **Save** (`Ctrl`+`S`) then writes *one* `.erisma` file with all items, their inputs, links and the project details (file name from the job no. or project name); **Open** restores the whole project. ↺ **Reset** on a project item also removes its links.

12. Linking calculations (calculation chain)

An input of one project item can take its value from a **result** of another item, so a change travels down the chain by itself — e.g. the axial force of the column becomes the load on the footing.

1. Open the receiving item (e.g. “Footing F1”) from the project list. Every numeric input now has a 🔗 button next to its label.
2. Click 🔗. Choose the **source calculation**; the list shows only its results whose unit is of the **same kind** as the input (force with force, moment with moment, …), with symbol, description and current value. Units are converted automatically (e.g. kN → MN).
3. Optionally type a **factor** — e.g. `1/1,35` to turn a design value N_{Ed} into a characteristic permanent load G — and tick **Use absolute value** for results that may be negative (e.g. hogging moments).
4. Press **Link**. The field becomes read-only and shows “🔗 from Column C1: N_Ed × 0,741”; **×** in that line removes the link.

Value used = factor × (|source result| if “absolute” is ticked, else source result). Before each run Erisma recalculates the whole project in dependency order (sources first), so the value is always current. If the source item is deleted or its result no longer exists, the field is marked red (“Link broken”) and the last value is kept. Links that form a loop (A → B → A) are refused with the message “Circular link between calculations” naming the items. Linked values are visible in the report as ordinary inputs, so the sheet remains checkable by hand.

13. Project report and company logo

📄 in the project header produces the whole project as one printable document:

- **Cover page** with your company logo, project name and the project details (company, job no., engineer, checker, date, National Annex).

- **Contents**: every item with its calculation, result (PASS / FAIL) and maximum utilisation.
- Every calculation sheet in full, in the order of the project list.

It opens in a new tab; print it with `Ctrl`+`P` or save it as PDF. **Logo:** in the project details press 🖼 **…** next to “Company logo” and choose an image (PNG, JPG, SVG…). It is reduced to at most 360 px to keep project files small, stored in the project file and printed on the cover and in the title block. “Remove logo” deletes it.

14. Quantities and cost estimate

Calculations of members that are built (footings, walls, beams, columns, slabs, piles, steel and timber members, strengthening…) end with a table **Quantities and cost estimate**: concrete, reinforcement, structural steel, timber, masonry, gabions, formwork, excavation, pile boring, ground anchors or FRP, each with quantity, unit price and cost. The total also appears in the status banner (💶 **≈ … €**; hover for the quantities). Section “Calculation reference” states for each calculation whether it has quantities.

- Quantities come from the actual geometry and the reinforcement chosen in the calculation (steel at 7850 kg/m³). Where the bar layout is not modelled in detail (e.g. walls, tanks) a typical reinforcement ratio is used and stated under the table.
- **Unit prices**: press 💶 **Unit prices** in the project details or ✎ on the cost chip. Defaults (€, supply and placing): concrete 130/m³, reinforcement 1,30/kg, structural steel 3,20/kg, timber 950/m³, masonry 160/m³, gabions 95/m³, formwork 25/m², excavation 12/m³, pile boring 70/m, anchors 90/m, FRP 120/m² per layer. Your prices are kept with the project and the project file; “Reset prices” returns to the defaults.

> The estimate is for comparing alternatives and preliminary budgets; it is not a bill of quantities (no laps, wastage, site overheads or VAT).

15. Automatic design — most economical solution

🎯 **Most economical** (above the form) searches for the **cheapest design that passes every check**. It is available for the calculations listed with “Automatic design” in the calculation reference, together with the variables it may change.

1. The dialog lists the design variables (e.g. footing B_x, B_y, h; beam depth, number and diameter of bars; steel section). Untick the ones that must stay as they are (e.g. a width fixed by the architect) and adjust **From / To / Step** (in the field's unit). Choice variables (sections, bar diameters) try all options.
2. Press **Search**. Every candidate is calculated in full with your other inputs, loads and National Annex. Ranking: first designs where all checks pass, then lowest cost (from the quantities and your unit prices), then lowest utilisation.
3. The result shows current and optimum values side by side (changed rows highlighted), cost and utilisation of both, number of trials, method and time. **Apply** puts the optimum into the form (`Ctrl`+`Z` undoes it).

Method: if the combinations are at most 2500 every one is tried (*exhaustive*, the true optimum on the grid). Otherwise *coordinate descent*: starting from your design, each variable in turn is run over its whole range and the best value kept, repeated (up to 5 rounds) until nothing improves — this gives a very good but not always the global optimum, so it is worth starting from a sensible design. A search stops after 40 s and returns the best found. If nothing passes in the ranges: "No passing design found — widen the ranges". The optimum is an ordinary design: check it and round it to practical dimensions if necessary.

16. Comparing alternative solutions

Erisma compares different solutions for the same element — e.g. footing 2,0 × 2,0 m against 2,4 × 2,4 m, or a cantilever RC wall against a gravity wall — by checks and by cost.

1. Open the calculation and enter the first solution.
2. Press ⚖ **Alternative** above the form. The current solution is kept in the project ("… – Solution A") and a copy opens with the name you give (default "… – Solution B").
3. Change the copy — other dimensions, other reinforcement, or open another calculation (e.g. a gravity wall) and add it with **+ Add to project**. Press ⚖ Alternative again for more solutions; a note above the form confirms each one.
4. Press ⚖ **Compare** in the project list on the left.

The table groups the project items by kind of calculation. In each group the solutions are sorted by cost and show result (PASS / FAIL), maximum utilisation U, quantities, cost and the **difference** from the cheapest solution that passes (e.g. "+230 € (+38%)"). The cheapest solution that passes all checks is marked ★; solutions that fail a check are greyed out and marked "not eligible". Costs come from the quantities and the unit prices (💶 in the project details), so set your own prices for a meaningful comparison. With fewer than two solutions in the project ⚖ Compare shows these steps instead of a table.

To compare *wall types*, give the items clear names ("Wall W1 – cantilever", "Wall W1 – gravity"): different calculation types are listed in separate groups, so read the costs of the groups side by side.

17. DXF drawings for CAD

For the calculations marked "DXF drawing" in the calculation reference the **DXF** button in the top bar downloads a drawing of the current design: plan and/or section with outline, reinforcement (bars and diameters, e.g. 5Ø16/200), soil line, dimensions and a title with element and project. It opens in AutoCAD and compatible CAD programs (BricsCAD, ZWCAD, progeCAD…) (format DXF R12, Greek code page, so Greek text is correct).

- **Units:** drawing units are millimetres at true size (1 unit = 1 mm); scale it at plotting.
- **Layers** (with colours): CONCRETE (7), REBAR (1 red), SOIL (34), DIM (3 green), TEXT (2 yellow), AXIS (8), STEEL (5 blue), HATCH (9) — so you can freeze or re-colour them to your office standard.

- The drawing is a starting point for the construction drawing; detailing (laps, anchorage lengths, bar schedules) is the designer's responsibility.

18. Trial and licence

Erisma can be used free of charge with all functions for **30 days** from the first start on a computer. The button in the top bar shows the state: "⏳ Trial: *n* days left", "✓ Licensed to *name*" or "⚠ Trial ended".

- **Activating:** click that button, paste the licence key you received (`ERISMA-…` ; spaces and line breaks are ignored) and press **Activate**. The key is checked on your computer (no internet needed) and kept in your user profile (`%APPDATA%\Erisma` on Windows, `~/.config/erisma` on Mac/Linux), so it survives updates of the program.
- **After the trial** without a key, calculations, reports, project reports, comparison, automatic design and DXF are blocked and the licence window opens; your inputs and project files are kept and everything works again as soon as a key is entered. The manual stays available.
- A key carries the licensee's name, the plan and possibly an expiry date. A key that is not accepted gets a clear reply ("not an Erisma licence key", "damaged — copy it again", "not valid", "expired on …"). When a key with an expiry date runs out the program falls back to the trial state. Keys are digitally signed (Ed25519) and cannot be altered or made up.

19. Keyboard shortcuts

Keys	Action
`Ctrl+K`	Search calculations
`Ctrl+S`	Save project file
`Ctrl+O`	Open project file
`Ctrl+P`	Print / PDF
`Ctrl+3`	Report ↔ 3D model
`Ctrl+Z / Ctrl+Y`	Undo / redo inputs
`Enter / Shift+Enter`	Next / previous field
`↑ / ↓ (Shift)`	Change the value (×10)
`F1`	This manual

20. National Annexes in Erisma

EN recommended values

EN 1990/1992/1997/1998 recommended values

Cyprus (CYS National Annexes)

NA to CYS EN 1990:2002, EN 1992-1-1:2004, EN 1997-1:2004, EN 1998-1:2004, EN 1998-5:2004 (approved 11/06/2010)

- Design Approach 2 is mandatory for STR/GEO (NA 2.9).
- Seismic zones: Zone 1 0.15 g, Zone 2 0.20 g, Zone 3 0.25 g (NA zonation map, 10 % in 50 years).

21. Calculation reference

ANALYSIS

Continuous beam analysis

Standard: Stiffness method

Linear elastic analysis of single or multi-span beams with any number of supports, point loads, UDLs and moments. Gives reactions, shear, moment and deflection diagrams.

Available tools: 3D model

Results that can be linked: L, E, I, M_max,sag, M_max,hog, V_max, δ_max

Method and assumptions

- Linear elastic stiffness (finite element) method, Euler–Bernoulli beam.
- Supports pinned or fixed at any position; loads downward positive.

Inputs

Input	Unit	Example	Range	Notes
Total length	m	10	0.1 – …	
Material		Steel (E = 210 GPa)	Steel (E = 210 GPa) / Concrete C25/30 (E = 31 GPa) / Timber C24 (E = 11 GPa) / Aluminium (E = 70 GPa)	
Second moment of area I	cm⁴	8356	0.001 – …	
Supports	table			Position [m], Type
Distributed loads	table			Start [m], End [m], w (down +) [kN/m]
Point loads	table			Position [m], P (down +) [kN]
Applied moments	table			Position [m], M (clockwise +) [kNm]

RC building — 3D analysis and design

Standard: EN 1998-1 / EN 1992-1-1

Give the grid, storeys, sizes and loads: 3D model, modal response spectrum analysis, drift and θ checks and the reinforcement of every beam and column.

Available tools: 3D model · quantities and cost

Results that can be linked: L_x × L_y, H, E_cm, E_eff, g_slab, G/A, α_cr, ψ_E, M, a_gR, a_g, F_b,X, F_b,Y, v ...

Method and assumptions

- 3D frame of Timoshenko beam elements with a rigid diaphragm per floor; slab loads to the beams by the 45° rule (exact fixed-end forces for trapezoids).
- Modal response spectrum analysis (CQC) in X and Y, accidental torsion ±0.05 L, E_X + 0.3 E_Y, drift d_r v ≤ α h and θ (EN 1998-1 4.4.2.2) per storey.
- Every beam and column is designed by the EC2/EC8 modules with the lightest passing bars; the critical members are printed in full.
- Slabs per panel (EC2, two-way), beam deflection by span/depth (EC2 7.4.2) and the foundation: beams on Winkler springs (EC2 + EC7 bearing, EN 1998-1 4.4.2.6 overstrength) or pad footings (EC7).

Inputs

Input	Unit	Example	Range	Notes
Spans along X	table			Span [m] — Grid lines 1, 2, 3... along X; one row per bay.
Spans along Y	table			Span [m] — Grid lines A, B, C... along Y; one row per bay.
Storeys (bottom up)	table			Storey height [m] — Floor-to-floor heights; the base is fixed.
Shear walls	table			Grid point, Direction, Length [m], Thickness [mm] — A wall replaces the column at that grid point (e.g. B1). Beams framing along the wall get rigid end zones of half its length.
Load cases and combinations		{}		Extra load cases (permanent, imposed, snow) with area, line and point loads, and user combinations — edited in Erisma Structure.

Plan layout		{}		Missing columns/beams, slab openings and member sizes — drawn in Erisma Structure.
Column side along X	mm	400	200 – 1500	
Column side along Y	mm	400	200 – 1500	
Beam width	mm	250	150 – 800	
Beam depth (incl. slab)	mm	500	250 – 1500	
Slab thickness	mm	180	100 – 400	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover to links	mm	30	15 – 60	
Finishes and partitions g	kN/m²	1.5	0 – 20	Floor finishes, screed, light partitions.
Imposed load q	kN/m²	2	0 – 20	EN 1991-1-1: dwellings 2.0, offices 3.0, assembly 4.0–5.0 kN/m².
Roof imposed load	kN/m²	2	0 – 20	
Facade walls on perimeter beams	kN/m	8	0 – 40	e.g. 3.6 kN/m² brick wall × 2.2 m = 8 kN/m. Not on the roof.
Use category (ψ factors)		A	A / B / C / D / E	EN 1990 Table A1.1: A dwellings, B offices, C assembly, D shopping, E storage.
Seismic design (EN 1998-1)		yes		
Reference ground		0.16	0 – 0.6	In g. Pick a zone with the buttons of

acceleration a_gR				the selected National Annex.
Ground type		B	A / B / C / D / E	
Importance class		II	I / II / III / IV	
Behaviour factor q		3.9	1 – 6	RC frame DCM multi-storey: 3.0·αu/α1 = 3.0·1.3 = 3.9.
φ for ψ_E = φ ψ_2 (mass)		1	0.5 – 1	EN 1998-1 Table 4.2: 1.0 roof, 0.8 correlated storeys, 0.5 independently occupied storeys.
Non-structural elements (drift limit)		Brittle non-structural elements (α = 0.005)	Brittle non-structural elements (α = 0.005) / Ductile non-structural elements (α = 0.0075) / No interfering non-structural elements (α = 0.010)	
Cracked stiffness factor		0.5	0.3 – 1	EN 1998-1 4.3.1(7): 50 % of the uncracked stiffness.
Apply EC8 regularity consequences to q		yes		Not regular in elevation: q × 0.8 (4.2.3.1(7)); torsionally flexible: q ≤ 2.0 (Table 5.1).
Number of modes		12	3 – 60	
P-Δ effects (geometric stiffness)		no		Include the destabilising effect of the gravity loads G + ψ2Q in the stiffness (all analyses).
Design the reinforcement of all members		yes		
Design the slabs (EN		yes		Every slab panel as a two-way slab,

1992-1-1)				continuous where a neighbour panel exists; bottom and top bars are chosen automatically.
Design the footings (EN 1997-1)		yes		A pad footing under every column and wall, sized for bearing, sliding and eccentricity; seismic actions with capacity design (EN 1998-1 4.4.2.6).
Foundation type		Raft (Winkler)	Raft (Winkler) / Foundation beams (Winkler) / Isolated pad footings	Raft: a plate under the whole building on elastic soil springs, analysed with finite elements together with the frame. Foundation beams along the grid lines on springs, or pad footings under a fixed base.
Raft thickness	mm	800	300 – 2000	
Raft overhang beyond the outer grid lines	m	1	0 – 3	
Raft mesh size	m	1	0.3 – 2	Finite-element size; smaller is more accurate but slower.
Foundation beam flange width B	m	1.2	0.4 – 4	
Foundation beam web width b_w	mm	500	250 – 1000	
Foundation beam depth h	mm	1200	500 – 2500	
Modulus of subgrade reaction k_s	MN/m³	40	5 – 300	Typical: loose sand 10–25, medium sand 25–60, dense sand 60–120, stiff clay 25–50 MN/m³ (or from a plate-load test / the geotechnical report).

Founding depth D	m	1.2	0.5 – 5	
Soil φ'k	°	30	15 – 45	
Soil c'k	kN/m²	5	0 – 100	
Soil γ	kN/m³	19	14 – 23	
Capacity design (EN 1998-1 DCM)		yes		ΣM_Rc ≥ 1.3 ΣM_Rb at the joints (4.4.2.3) and shears from the flexural capacities (5.4.2.2, 5.4.2.3).

Checks performed (worked example)

Check	Clause	U (example)
α_cr ≥ 10 (first-order analysis sufficient)	EN 1993-1-1 Eq. 5.1	0.06
Σ effective mass in X ≥ 90 %	EN 1998-1 4.3.3.3.1(3)	0.9
Σ effective mass in Y ≥ 90 %	EN 1998-1 4.3.3.3.1(3)	0.9
Storey 1 drift X: d_r ν ≤ α h	EN 1998-1 4.4.3.2	0.22
Storey 1 drift Y: d_r ν ≤ α h	EN 1998-1 4.4.3.2	0.23
Storey 2 drift X: d_r ν ≤ α h	EN 1998-1 4.4.3.2	0.31
Storey 2 drift Y: d_r ν ≤ α h	EN 1998-1 4.4.3.2	0.34
Storey 3 drift X: d_r ν ≤ α h	EN 1998-1 4.4.3.2	0.29
Storey 3 drift Y: d_r ν ≤ α h	EN 1998-1 4.4.3.2	0.32
Second-order sensitivity θ ≤ 0.30	EN 1998-1 4.4.2.2(4)	0.09
Storey 1: ΣM_Rc ≥ 1.3 ΣM_Rb (joint C2@1)	EN 1998-1 4.4.2.3(4)	0.63
Storey 2: ΣM_Rc ≥ 1.3 ΣM_Rb (joint C2@2)	EN 1998-1 4.4.2.3(4)	0.79
Raft bottom face x: A_s,prov ≥ A_s,req	EN 1992-1-1 6.1	0.69
Raft bottom face y: A_s,prov ≥ A_s,req	EN 1992-1-1 6.1	0.69
Raft top face x: A_s,prov ≥ A_s,req	EN 1992-1-1 6.1	0.63
Raft top face y: A_s,prov ≥ A_s,req	EN 1992-1-1 6.1	0.65
Raft shear away from the columns: V_Ed ≤ V_Rd,c	EN 1992-1-1 6.2.2	0.69
Raft punching at column A4	EN 1992-1-1 6.4	0.7
Soil bearing under the raft (DA2)	EN 1997-1 6.5.2	0.04
Soil bearing under the raft (G+ψ2Q+ΩE)	EN 1997-1 6.5.2	0.04
Relative rotation between columns B3–C3 ≤ 1/500	EN 1997-1 Annex H	0.01
Total settlement (G + ψ_2Q) ≤ 50 mm	EN 1997-1 Annex H	0.03
Columns, storey 1: max utilisation (C1.A1)		0.78
Columns, storey 2: max utilisation (C2.A1)		0.78

Columns, storey 3: max utilisation (C3.C2)		0.8
Beams, storey 1: max utilisation (B1.A2-A3)		0.99
Beams, storey 2: max utilisation (B2.B1-C1)		0.98
Beams, storey 3: max utilisation (B3.A3-B3)		0.97
Walls (critical region at the base), storey 1: max utilisation (W1.B1)		0.94
Slabs, storey 1: max utilisation (S1.B2)		0.93
Slabs, storey 2: max utilisation (S2.B2)		0.93
Slabs, storey 3: max utilisation (S3.B2)		0.93

CONCRETE

Pile cap (strut-and-tie)

Standard: EN 1992-1-1 §6.5, §9.8.1

Rigid pile cap on 2, 3 or 4 piles under a column: pile loads with moments, strut-and-tie model, tie reinforcement, strut angle and node stresses at the column and the piles. Interactive 3D model.

Available tools: 3D model · quantities and cost

Results that can be linked: d, P_max, θ, T_Ed, A_s, σ_c,col, σ_c,pile

Method and assumptions

- Rigid cap: pile loads from N and M; strut-and-tie with z = 0.9d, ties along the pile lines; node stresses by Eq. 6.60/6.61.

Inputs

Input	Unit	Example	Range	Notes
Number of piles		4	2 / 3 / 4	
Pile diameter	mm	600	250 – 1500	
Pile spacing (centre to centre)	mm	1800	600 – …	Usually ≥ 3 D.
Cap depth h	mm	1000	400 – 3000	
Column size (square)	mm	500	200 – …	
Edge distance beyond the piles	mm	300	150 – …	
Design axial load N_Ed	kN	3500	0 – …	
Design moment M_x,Ed	kNm	150		
Design moment M_y,Ed	kNm	0		
Concrete class		C30/37	C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	

Cover to the tie bars	mm	75	50 – …	
Tie bars per tie		7	2 – 30	
Tie bar Ø		20	16 / 20 / 25 / 32	

Checks performed (worked example)

Check	Clause	U (example)
Strut angle θ ≥ 26.5° (cot θ ≤ 2)	6.5.2 / 6.2.3	0.72
Tie: T_Ed ≤ A_s f_yd	6.5.3	0.91
Column node (C-C-C): σ ≤ 1.0 ν' f_cd	Eq. 6.60	0.8
Pile node (C-C-T): σ ≤ 0.85 ν' f_cd	Eq. 6.61	0.22
Edge distance beyond the pile face ≥ 150 mm	good practice	0.5

RC beam section design

Standard: EN 1992-1-1:2004

Rectangular section with tension and compression reinforcement: flexure (stress block, strain compatibility), minimum/maximum reinforcement and shear with variable strut inclination.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Overall depth h, Number of tension bars, Tension bar diameter)

Results that can be linked: b, h, c_nom, d, f_ck, f_cd, f_ctm, f_yk, f_yd, M_Ed, μ, μ_lim, ξ, z …

Method and assumptions

- Rectangular stress block (λ = 0.8, η = 1.0), ε_cu3 = 3.5 ‰, x/d ≤ 0.45 for ductility.
- Shear with variable strut inclination 1 ≤ cot θ ≤ 2.5 (6.2.3).

Inputs

Input	Unit	Example	Range	Notes
Width b	mm	300	100 – …	
Overall depth h	mm	500	150 – …	
Nominal cover to links	mm	30	10 – …	
Concrete class		C25/30	7 …	
Reinforcement fyk	N/mm²	500	200 – 600	
γc		1.5	1 – …	
γs		1.15	1 – …	
αcc		1	0.8 – 1	1.0 recommended (Greek NA); 0.85 UK NA.

Design moment MEd	kNm	140	0 – …	
Design shear VEd	kN	120	0 – …	
Number of tension bars		4	2 – …	
Tension bar diameter		16	8 …	
Number of compression (top) bars		2	2 – …	
Compression bar diameter		12	8 …	
Link diameter		8	8 / 10 / 12 / 14	
Link legs		2	2 – …	
Link spacing	mm	200	50 – …	

Checks performed (worked example)

Check	Clause	U (example)
Tension reinforcement: A_s,prov ≥ A_s,req	6.1	0.95
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1(1)	0.23
Maximum reinforcement: A_s + A_s2 ≤ A_s,max	9.2.1.1(3)	0.17
Bar spacing: s_clear ≥ max(φ, d_g + 5, 20)	8.2(2)	0.47
Moment resistance: M_Ed ≤ M_Rd	6.1	0.96
Concrete strut: V_Ed ≤ V_Rd,max	6.2.3(3)	0.32
Shear reinforcement: V_Ed ≤ V_Rd,s	6.2.3(3)	0.54
Minimum links: A_sw/(s b) ≥ ρ_w,min	9.2.2(5)	0.48
Link spacing: s ≤ 0.75d	9.2.2(6)	0.59

RC column

Standard: EN 1992-1-1 §5.8, EN 1998-1 §5.4

Rectangular column: imperfections, slenderness, second-order effects (nominal curvature), biaxial bending (5.8.9), shear, detailing and EC8 DCM rules. Interactive 3D with reinforcement.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Width b, Depth h (bending about the strong axis), Longitudinal bars Ø)

Results that can be linked: b × h, f_cd, f_yd, A_s, N_Ed, ω, n, N_Rd, l_0,h, λ_h, λ_lim,h, e_i,h, M_Ed,h, M_Rd,h …

Method and assumptions

- Second-order effects by the nominal curvature method (5.8.8); biaxial by Eq. 5.39.
- M_Rd by strain compatibility for the actual axial force; EC8 DCM rules optional.

Inputs

Input	Unit	Example	Range	Notes
Width b	mm	400	200 – …	
Depth h (bending about the strong axis)	mm	400	200 – …	
Cover to links	mm	30	15 – …	
Concrete class		C25/30	7 …	
Reinforcement fyk	N/mm²	500	400 – 600	
Longitudinal bars Ø		16	9 …	
Bars along b (each face, incl. corners)		4	2 – …	
Bars along h (each face, incl. corners)		4	2 – …	
Links Ø		8	8 / 10 / 12	
Link spacing	mm	100	50 – …	
Clear height	m	3	0.5 – …	
Effective length factor (bending in h)		1	0.5 – 2.5	0.5 fixed–fixed, 0.7 fixed–pinned, 1.0 pinned–pinned, 2.0 cantilever (or EN 1992-1-1 Eq. 5.15/5.16).
Effective length factor (bending in b)		1	0.5 – 2.5	
Braced column		yes		Unbraced columns use r_m = 1.0 (C = 0.7).
Axial force N_Ed (compression +)	kN	1200	0 – …	
M_Ed (h) at top	kNm	60		First-order end moments; same sign = single curvature.
M_Ed (h) at bottom	kNm	-30		
M_Ed (b) at top	kNm	20		
M_Ed (b) at bottom	kNm	-10		
Shear V_Ed	kN	60	0 – …	
Effective creep ratio φ_ef		0.8	0 – 3	
Primary seismic column (EN 1998-1 DCM)		yes		

Checks performed (worked example)

Check	Clause	U (example)
Axial: N_Ed ≤ N_Rd	6.1	0.32

Bending in h: M_Ed ≤ M_Rd	6.1	0.28
Bending in b: M_Ed ≤ M_Rd	6.1	0.12
Biaxial: (M_Ed,h/M_Rd,h)^a + (M_Ed,b/M_Rd,b)^a ≤ 1	EN 1992-1-1 Eq. 5.39	0.31
Shear: V_Ed ≤ max(V_Rd,c, min(V_Rd,s, V_Rd,max))	6.2	0.17
A_s ≥ A_s,min = max(0.10 N_Ed/f_yd, 0.002 A_c)	9.5.2(2)	0.13
A_s ≤ 0.04 A_c	9.5.2(3)	0.38
Link spacing ≤ min(20φ, b_min, 400)	9.5.3(3)	0.31
Axial load ratio ν_d ≤ 0.65	EN 1998-1 5.4.3.2.1(3)	0.69
ρ_l ≥ 1 %	5.4.3.2.2(1)	0.66
ρ_l ≤ 4 %	5.4.3.2.2(1)	0.38
Critical region links: s ≤ min(b_0/2, 175, 8d_bL)	5.4.3.2.2(11)	0.78

RC shear wall (EC8 DCM)

Standard: EN 1998-1 §5.4, EN 1992-1-1

Ductile cantilever wall of ductility class DCM: geometry, normalised axial load, critical region, flexure with boundary elements, shear with the amplification ε = 1.5, web reinforcement and confinement of the boundary elements (Eq. 5.20). Interactive 3D model.

Available tools: 3D model · quantities and cost

Results that can be linked: h_cr, ν_d, A_s,be, M_Rd, V_Ed, V_Rd,max, V_Rd,s, μ_φ, ω_wd, α

Method and assumptions

- DCM wall: ν_d ≤ 0.4, h_cr (Eq. 5.19), M_Rd by strain compatibility with web bars, V_Ed = 1.5 V'_Ed, confinement of boundary elements by Eq. 5.20 with μ_φ = 2q_0 − 1.

Inputs

Input	Unit	Example	Range	Notes
Wall length l_w	m	4	1 – 12	
Wall thickness b_w	mm	250	150 – 600	
Clear storey height h_s	m	3	2 – 6	
Total wall height H_w	m	18	3 – 120	
Number of storeys		6	1 – 40	
Axial force N_Ed (seismic, compression +)	kN	2500	0 – …	
Moment M_Ed	kNm	9000	0 – …	
Shear V'_Ed (from the analysis)	kN	700	0 – …	

Basic behaviour factor q_0		3	1.5 – 4.5	DCM wall systems: q_0 = 3.0 (Table 5.1).
Concrete class		C30/37	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover to the hoops	mm	30	20 – …	
Boundary element length l_c	mm	700	200 – …	
Bars in each boundary element		8	4 – 30	
Boundary bar Ø		20	10 / 12 / 14 / 16 / 20 / 25	
Boundary hoop Ø		10	8 / 10 / 12	
Boundary hoop spacing	mm	85	50 – 250	
Web vertical bars Ø (each face)		10	10 / 12 / 14 / 16 / 20 / 25	
Web vertical spacing	mm	200	75 – 400	
Web horizontal bars Ø (each face)		12	10 / 12 / 14 / 16 / 20 / 25	
Web horizontal spacing	mm	150	75 – 400	

Checks performed (worked example)

Check	Clause	U (example)
Thickness b_w ≥ max(150 mm, h_s/20)	5.4.1.2.3(1)	0.6
Normalised axial load v_d ≤ 0.4	5.4.3.4.1(2)	0.31
Flexure: M_Ed ≤ M_Rd	5.4.3.4.1	0.98
Boundary element ρ ≥ 0.5 %	5.4.3.4.2(8)	0.35
Boundary length l_c ≥ max(0.15 l_w, 1.5 b_w)	5.4.3.4.2(6)	0.86
Web crushing: V_Ed ≤ V_Rd,max	6.2.3	0.25
Shear: V_Ed ≤ V_Rd,s	6.2.3	0.5
Web vertical ρ_v ≥ 0.2 %	EN 1992-1-1 9.6.2	0.64
Web horizontal ρ_h ≥ max(0.25 ρ_v, 0.1 %)	EN 1992-1-1 9.6.3	0.17
Confinement: α ω_wd ≥ 30 μ_φ (v_d + ω_v) ε_sy,d b_c/b_0 − 0.035	Eq. 5.20	0.34
Boundary ω_wd ≥ 0.08	5.4.3.4.2(4)	0.18
Hoop spacing ≤ min(b_0/2, 175, 8 d_bL)	5.4.3.2.2(11)	0.94

RC slab

Standard: EN 1992-1-1

One-way continuous slabs with pattern loading, or two-way rectangular panels with any edge condition (plate analysis); bottom/top reinforcement, shear, deflection (span/depth) and punching at a column.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Slab thickness h, Bottom bar spacing, Top bar spacing)

Results that can be linked: g_k, q_k, p_d, m_x,span, m_y,span, m_x,sup, m_y,sup, μ, μ_lim, ξ, z, A_s,req, A_s,min, A_s,max ...

Method and assumptions

- Two-way panels: Kirchhoff plate by finite differences with any edge conditions (verified against Timoshenko's tables).
- One-way continuous slabs: pattern loading (all / odd / even spans).

Inputs

Input	Unit	Example	Range	Notes
Slab type		Two-way panel	Two-way panel / One-way continuous	
Short span l_x	m	4.5	1 – ...	
Long span l_y (two-way)	m	6	1 – ...	
Number of equal spans (one-way)		3	1 – 8	
Edge x = 0		Fixed / continuous	Simply supported / Fixed / continuous	
Edge x = l_x		Simply supported	Simply supported / Fixed / continuous	
Edge y = 0		Fixed / continuous	Simply supported / Fixed / continuous	
Edge y = l_y		Simply supported	Simply supported / Fixed / continuous	
Slab thickness h	mm	180	100 – ...	
Cover	mm	25	15 – ...	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Superimposed permanent load (excl. self weight)	kN/m²	2	0 – ...	Finishes, screed, partitions.

Imposed load	kN/m²	2	0 – …	EN 1991-1-1: residential 1.5–2.0, offices 3.0, balconies 4.0.
Bottom bars Ø		10	8 / 10 / 12 / 14 / 16	
Bottom bar spacing	mm	150	75 – 400	
Top bars Ø (supports)		12	8 / 10 / 12 / 14 / 16	
Top bar spacing	mm	150	75 – 400	
Check punching at a column (flat slab)		no		
Column size c1	mm	400	150 – …	
Column size c2	mm	400	150 – …	
Column reaction V_Ed	kN	450	0 – …	
Column position		Internal	Internal / Edge / Corner	β = 1.15 / 1.4 / 1.5 (EN 1992-1-1 Fig. 6.21N).

Checks performed (worked example)

Check	Clause	U (example)
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.35
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.38
Bending: M_Ed ≤ M_Rd	6.1	0.34
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.23
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.36
Bending: M_Ed ≤ M_Rd	6.1	0.23
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.49
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.26
Bending: M_Ed ≤ M_Rd	6.1	0.5
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.4
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.26
Bending: M_Ed ≤ M_Rd	6.1	0.41
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.41
Span/depth: l/d ≤ (l/d)_lim	7.4.2	0.59

RC stair flight

Standard: EN 1992-1-1, EN 1991-1-1

Straight flight with a waist slab spanning between landings or beams: geometry rules (2R + G), self weight of waist and steps, bending, shear, minimum reinforcement and span/depth. Interactive 3D model.

Available tools: 3D model · quantities and cost

Results that can be linked: R, G, α, g_sw, g_k, p_d, M_Ed, V_Ed, d, μ, μ_lim, ξ, z, A_s,req …

Method and assumptions

- Waist slab as a one-way slab on the horizontal span; self weight of waist and steps on plan; comfort rule 2R + G.

Inputs

Input	Unit	Example	Range	Notes
Horizontal span (support to support)	m	3.08	1 – 8	
Total rise of the flight	m	2.04	0.3 – 4	
Number of risers		12	2 – 25	
Flight width	m	1.2	0.6 – 4	
Waist thickness (perpendicular to the slope)	mm	200	100 – 350	
Support condition		Simply supported	Simply supported / Continuous with landings	Continuous: M = wL²/10 (end spans with landing slabs).
Finishes (on plan)	kN/m²	1.5	0 – …	
Imposed load	kN/m²	3	0 – …	EN 1991-1-1 category A stairs: 2.0–4.0 kN/m² (national annex).
Concrete class		C25/30	C20/25 / C25/30 / C30/37	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover	mm	25	15 – …	
Main bars Ø		12	8 / 10 / 12 / 14 / 16	
Main bar spacing	mm	150	75 – 300	

Checks performed (worked example)

Check	Clause	U (example)

Comfort: 2R + G ≤ 650 mm	building regulations (600–650)	0.95
Comfort: 2R + G ≥ 580 mm	building regulations (600–650)	0.94
Riser R ≤ 190 mm	building regulations	0.89
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.39
Minimum reinforcement	9.3.1.1	0.3
Bending: M_Ed ≤ M_Rd	6.1	0.4
Bar spacing ≤ min(3t, 400)	9.3.1.1(3)	0.38
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.25
Span/depth: l/d ≤ (l/d)_lim (horizontal span)	7.4.2	0.91

Water tank / pool

Standard: EN 1992-3, EN 1992-1-1, EN 1997-1

Rectangular reinforced concrete tank or swimming pool: wall bending for the full (water test) and empty (backfill + groundwater) cases, crack width to EN 1992-3 tightness class 1, minimum thickness and reinforcement, and flotation (UPL) of the empty tank. Interactive 3D.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Wall thickness, Spacing, inside face, Spacing, outside face)

Results that can be linked: H, h_w, t, w_k1, K_0, M_Ed,base, M_qp,base, d, M_Ed, μ, μ_lim, ξ, z, A_s,req …

Method and assumptions

- Long walls as vertical strips fixed into the base, free or pinned at the top.
- Crack width by EN 1992-1-1 7.3.4 (quasi-permanent) against w_k1 of tightness class 1.
- Thermal/early-age restraint cracking is not included.

Inputs

Input	Unit	Example	Range	Notes
Internal length	m	10	1 – …	
Internal width	m	5	1 – …	
Wall height (above base)	m	2.5	0.5 – 12	
Maximum water depth	m	2.3	0.1 – …	
Wall thickness	mm	250	120 – 1000	
Base slab thickness	mm	300	150 – 1500	
Base projection beyond the walls	m	0.3	0 – …	Soil resting on the projection helps against flotation.

Wall support at the top		Free (open tank / pool)	Free (open tank / pool) / Pinned (roof slab)	
Backfill height outside (above base)	m	2.5	0 – …	
Backfill γ	kN/m³	19	10 – …	
Backfill φ'k	°	30	20 – 45	
Surcharge on the backfill	kN/m²	5	0 – …	
Groundwater outside (above base underside)	m	1	0 – …	Highest expected level — governs flotation of the empty tank.
Concrete class		C30/37	C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover to bars	mm	40	25 – 75	
Vertical bars, inside face Ø		12	10 / 12 / 14 / 16 / 20	
Spacing, inside face	mm	125	75 – 300	
Vertical bars, outside face Ø		12	10 / 12 / 14 / 16 / 20	
Spacing, outside face	mm	125	75 – 300	

Checks performed (worked example)

Check	Clause	U (example)
Minimum wall thickness (EN 1992-3 9.11.1, classes 1–2)	EN 1992-3 9.11.1(102)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.31
Bending: M_Ed ≤ M_Rd	6.1	0.31
A_s ≥ A_s,min = k_c k f_ct,eff A_ct / f_yk (k_c = 0.4)	EN 1992-1-1 7.3.2	0.32
Crack width inside: w_k ≤ w_k1	EN 1992-3 7.3.1	0.56
Compression zone x ≥ x_min = min(50 mm, 0.2t) (no through-cracks)	EN 1992-3 7.3.1(112)	0.81
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.61
Bending: M_Ed ≤ M_Rd	6.1	0.61
A_s ≥ A_s,min = k_c k f_ct,eff A_ct / f_yk (k_c = 0.4)	EN 1992-1-1 7.3.2	0.32
Crack width outside: w_k ≤ w_k1	EN 1992-3 7.3.1	0.78
Compression zone x ≥ x_min = min(50 mm, 0.2t) (no through-cracks)	EN 1992-3 7.3.1(112)	0.81
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.46

UPL: γ_G,dst U ≤ γ_G,stb G (1 / 0.9)	EN 1997-1 Eq. 2.8	0.5

EXISTING BUILDINGS

Assessment and strengthening of RC members

Standard: EN 1998-3 Annex A

Existing RC column or beam: knowledge level and confidence factor, chord rotation at yield and ultimate (A.10a, A.1), cyclic shear resistance (A.12), checks for the limit states NC, SD and DL, and strengthening with an RC jacket or FRP wrapping. Interactive 3D model.

Available tools: 3D model · quantities and cost · automatic design (Jacket thickness, Jacket hoop spacing)

Results that can be linked: CF, f_c, f_y, f_c,V, f_yw,V, φ_y, M_y, a_v, θ_y, θ_um, θ_SD, V_R, V_My, b* × h* ...

Method and assumptions

- EN 1998-3 Annex A: θ_y (A.10a), θ_um (A.1), V_R (A.12); mean strengths ÷ CF, brittle shear also ÷ γ (Cyprus NA 2.3).
- NC: θ_E ≤ θ_um, SD: θ_E ≤ ¾θ_um, DL: θ_E ≤ θ_y. RC jacket by the monolithic factors of A.4.2.2, FRP by A.4.4. The Greek KANEPE is not implemented.

Inputs

Input	Unit	Example	Range	Notes
Knowledge level		KL2 — normal knowledge	KL1 — limited knowledge / KL2 — normal knowledge / KL3 — full knowledge	KL1: simulated design + limited in-situ tests; KL2: extended inspection and testing; KL3: comprehensive (Table 3.1).
Primary seismic element		yes		γ_el = 1.5 (deformation) and 1.15 (shear) for primary elements, 1.0 for secondary.
Detailed for earthquake resistance		no		Off for older buildings (no seismic detailing): θ_um is divided by 1.2 (A.3.2.2).
Ribbed (deformed) bars		yes		
Width b	mm	300	150 – ...	
Depth h (in the direction of bending)	mm	400	150 – ...	
Cover to bar centre	mm	35	15 – ...	

Bars along b (each face, incl. corners)		3	2 – …	
Bars along h (each face, incl. corners)		3	2 – …	
Longitudinal bar Ø	mm	16	8 – 32	
Hoop Ø	mm	8	6 – 14	
Hoop spacing	mm	200	50 – 400	
Hoop legs parallel to the shear		2	2 – 6	
Concrete mean strength f_cm (tests)	N/mm²	16	8 – 60	
Steel mean yield strength f_ym	N/mm²	450	220 – 700	
Shear span L_V = M/V	m	1.5	0.3 – …	Half the clear height for a column in double curvature.
Axial force in the seismic situation	kN	600	0 – …	Compression positive (G + ψ2Q).
Chord rotation demand θ_E — NC		0.02	0 – 0.2	rad, from the analysis for the NC seismic action.
Chord rotation demand θ_E — SD		0.012	0 – 0.2	
Chord rotation demand θ_E — DL		0.004	0 – 0.2	
Shear demand V_E	kN	150	0 – …	From capacity design: ΣM_y / L_cl of the member, or from the analysis.
Strengthening		RC jacket	None (assessment only) / RC jacket / FRP wrapping	
Jacket thickness	mm	100	60 – 250	≥ 75 mm cast, ≥ 50 mm shotcrete.
Jacket concrete mean strength	N/mm²	33	20 – 60	

Jacket bars along each face (incl. corners)		3	2 – …	
Jacket bar Ø	mm	16	10 – 32	
Jacket hoop Ø	mm	10	8 – 14	
Jacket hoop spacing	mm	100	50 – 200	
Jacket steel mean yield strength	N/mm²	550	400 – 700	
FRP layers		2	1 – 10	
Thickness per layer	mm	0.17	0.05 – 2	
FRP modulus E_f	GPa	230	20 – 650	
FRP ultimate strain ε_u,f		0.015	0.003 – 0.05	
Corner radius	mm	25	10 – 60	Round the corners before wrapping (≥ 20–25 mm).

Checks performed (worked example)

Check	Clause	U (example)
NC (with RC jacket): θ_E ≤ θ_um	EN 1998-3 A.3.2.2	0.72
SD (with RC jacket): θ_E ≤ ¾ θ_um	EN 1998-3 A.3.2.3	0.58
DL (with RC jacket): θ_E ≤ θ_y	EN 1998-3 A.3.2.4	0.62
Shear (with RC jacket): V_E ≤ V_R	EN 1998-3 A.3.3.1	0.44

FIRE

Fire resistance — RC beam / slab

Standard: EN 1992-1-2 §5 (tabulated data)

Tabulated data for simply supported beams (Table 5.5) and slabs (Table 5.8): minimum width / thickness and axis distance of the reinforcement for the required fire resistance.

Available tools: 3D model

Results that can be linked: a, a_req, a_sd

Method and assumptions

- Tabulated data: Table 5.5 (simply supported beams, interpolated in b, corner bars +10 mm) and Table 5.8 (slabs).

Inputs

Input	Unit	Example	Range	Notes
Member		Simply supported beam	Simply supported beam / One-way slab / Two-way slab (l_y/l_x ≤ 1.5) / Two-way slab (1.5 < l_y/l_x ≤ 2)	
Required fire resistance		R90	R30 / R60 / R90 / R120 / R180 / R240	
Beam width b / slab thickness h	mm	300	60 – …	
Nominal cover to the links (beam) / bars (slab)	mm	35	10 – …	
Link Ø (beam)	mm	8	0 – 16	
Main bar Ø	mm	20	8 – 40	
One layer of bars (beam: corner bars need a + 10 mm)		yes		

Checks performed (worked example)

Check	Clause	U (example)
Beam width b ≥ b_min (R90)	EN 1992-1-2 Table 5.5	0.5
Axis distance a ≥ a_req (R90)	EN 1992-1-2 5.6	0.94

Fire resistance — steel member

Standard: EN 1993-1-2, EN 1991-1-2

Steel I/H member in the standard fire: degree of utilisation, critical temperature, section factor with shadow effect, heating of the unprotected or protected member and the fire resistance time against the requirement.

Available tools: 3D model

Results that can be linked: A_m/V, [A_m/V]_b, k_sh, μ_0, θ_a,cr, A_p/V, λ_p, d_p, t_fi,d

Method and assumptions

- Critical temperature Eq. 4.22; heating to ISO 834 by Eq. 4.25 (unprotected, k_sh) or Eq. 4.27 (protected, product data λ_p, d_p, c_p, ρ_p).

Inputs

Input	Unit	Example	Range	Notes
Section		HEB 240	40 …	
Fire exposure		3 sides (beam under a slab)	4 sides (column) / 3 sides (beam under a slab)	
Utilisation at normal temperature E_d / R_d		0.8	0.05 – 1	From the cold design (e.g. the steel beam calculation).
Reduction factor for fire η_fi		0.65	0.2 – 1	E_fi,d = η_fi E_d; EN 1993-1-2 2.4.2(3): 0.65 recommended (0.7 for storage areas).
Required fire resistance		R60	R30 / R60 / R90 / R120 / R180 / R240	
Fire protection		Board / spray (λ_p, d_p)	None / Board / spray (λ_p, d_p)	
Thermal conductivity λ_p		0.12	0.02 – 1	W/mK — from the product data.
Protection thickness d_p	mm	20	2 – 100	
Specific heat c_p		1200	100 – 3000	J/kgK
Density ρ_p	kN/m³	3	0.5 – 20	e.g. 3 kN/m³ ≈ 300 kg/m³

Checks performed (worked example)

Check	Clause	U (example)
Fire resistance ≥ R60	EN 1993-1-2 2.4.2	0.36

Fire resistance — timber beam

Standard: EN 1995-1-2 §4.2.2

Reduced cross-section method: charring depth, effective section, strength in fire (k_fi, k_mod,fi = 1) and bending, lateral buckling and shear under the fire load combination.

Available tools: 3D model

Results that can be linked: β_n, d_char,n, d_ef, b_fi, h_fi, k_fi, f_m,d,fi, w_fi, σ_m,fi, k_crit,fi

Method and assumptions

- Reduced cross-section: d_ef = β_n t + 7 mm; f_d,fi = k_fi f_k (k_mod,fi = γ_M,fi = 1).

Inputs

Input	Unit	Example	Range	Notes
Strength class		GL24h	8 …	
Width b	mm	160	60 – …	
Depth h	mm	480	100 – …	
Span L	m	6	0.5 – …	
Exposure		3 sides (top protected)	3 sides (top protected) / 4 sides	
Permanent load g_k (incl. self weight)	kN/m	3	0 – …	
Variable load q_k	kN/m	3	0 – …	
ψ for the fire situation		0.3	0 – 1	EN 1991-1-2 4.3.1: ψ2,1 recommended (or ψ1,1 per the National Annex).
Required fire resistance		R30	R30 / R60 / R90 / R120	

Checks performed (worked example)

Check	Clause	U (example)
Bending in fire: σ_m,fi ≤ k_crit f_m,d,fi	EN 1995-1-2 4.2.2	0.21
Shear in fire: τ_fi ≤ k_fi f_v,k	EN 1995-1-2 4.2.2	0.14

GEOTECHNICAL

Berlin wall (soldier pile wall)

Standard: EN 1997-1, EN 1993-1-1, EN 1995-1-1, EN 1998-5

Steel soldier piles with timber lagging, cantilever or with one anchor row: active/passive earth pressure with spatial passive resistance, embedment (Blum / free earth support), pile bending and shear (EC3), lagging (EC5), anchor force, seismic case and indicative deflection. Interactive 3D model.

Available tools: 3D model · quantities and cost · automatic design (Pile section, Provided embedment below excavation t)

Results that can be linked: H, a, b, z_A, φ'_k, c'_k, γ, q_k, f, b_p, φ'_d, K_ah, K_ph, t_0 …

Method and assumptions

- Blum (cantilever) or free earth support (one anchor row); spatial passive resistance below the excavation (f·b ≤ a).

Inputs

Input	Unit	Example	Range	Notes
Excavation depth H	m	5	1 – 15	
Pile spacing a (centre to centre)	m	2	0.5 – 4	Typical 1.5–3.0 m.
Provided embedment below excavation t	m	3.5	0.5 – …	
Support system		Single anchor row	Cantilever / Single anchor row	
Anchor depth below top	m	1.5	0.3 – …	
Anchor inclination below horizontal	°	20	0 – 45	
Anchor design resistance R_a,d (per anchor)	kN	450	1 – …	From anchor design / suitability tests (EN 1997-1 §8).
Number of piles shown in 3D		6	2 – 12	
Pile section		HEB 240	40 …	
Steel grade		S355	S235 / S275 / S355	
Lagging board thickness	mm	80	30 – 200	
Lagging pressure factor (arching)		0.6	0.3 – 1	Arching reduces the pressure on the boards: ~0.6 in competent soils, 1.0 in soft clay or loose sand.
Soil φ'k	°	32	20 – 45	
Soil c'k	kN/m²	0	0 – …	
Soil γ	kN/m³	19	10 – …	
Active wall friction δa/φ'		0.667	0 – 0.667	
Passive wall friction δp/φ'		0.333	0 – 0.5	Keep ≤ 0.5: Coulomb overestimates passive resistance.
Surcharge behind the wall (variable)	kN/m²	10	0 – …	

Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	
Displacement factor r		1.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: use 1.0 for anchored / propped walls.
ψ2 for surcharge		0.3	0 – 1	

Checks performed (worked example)

Check	Clause	U (example)
DA2: t_prov ≥ t_req	EN 1997-1 9.7.4	0.62
Seismic: t_prov ≥ t_req	EN 1997-1 9.7.4	0.8
Pile bending: M_Ed ≤ M_c,Rd	EN 1993-1-1 6.2.5	0.42
Pile shear: V_Ed ≤ V_pl,Rd	EN 1993-1-1 6.2.6	0.19
Embedment: t_prov ≥ t_req	EN 1997-1 9.7.4	0.8
Anchor: A_d ≤ R_a,d	EN 1997-1 8.5	0.39
Lagging: t_prov ≥ t_req	EN 1995-1-1 6.1	0.69

Cantilever retaining wall

Standard: EN 1997-1, EN 1998-5, EN 1992-1-1

Reinforced concrete cantilever wall: active earth pressure (Coulomb), surcharge, groundwater, seismic thrust (Mononobe-Okabe), sliding, overturning, bearing (Annex D), and EC2 design of stem, heel and toe. Includes an interactive 3D model.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Toe length, Heel length (behind stem), Base thickness, Stem thickness at base)

Results that can be linked: H_stem, t_top, t_bot, t_b, b_toe, b_heel, B, β, h_v, q_k, φ'_d, K_a, φ'_f,d, tan δ_d ...

Method and assumptions

- Coulomb active pressure on the virtual back through the heel; Mononobe-Okabe in the seismic case (±k_v); bearing by Annex D with inclination factors.
- Passive resistance in front is ignored except at a shear key.

Inputs

Input	Unit	Example	Range	Notes
Stem height above base	m	4	0.5 – 15	
Stem thickness at top	m	0.25	0.15 – …	
Stem thickness at base	m	0.4	0.15 – …	
Base thickness	m	0.5	0.2 – …	
Toe length	m	1	0 – …	
Heel length (behind stem)	m	2.2	0.2 – …	
Soil depth over toe	m	0.5	0 – …	Depth of soil in front of the wall above the top of the base.
Wall length (3D view)	m	8	1 – …	
Stepped stem (thicker lower part)		no		Thicken the lower part of the stem where the moments are largest.
Height of the step above the base	m	1.5	0.2 – …	
Stem thickness below the step	m	0.55	0.2 – …	Above the step the stem tapers from 'thickness at base' to 'thickness at top'.
Shear key under the base		no		A downstand under the base that mobilises passive resistance against sliding.
Key position (front face, from the toe)	m	1	0 – …	Usually under the stem or just behind it.
Key width	m	0.4	0.2 – …	
Key depth below the base	m	0.5	0.2 – 2	
Backfill slope β	°	0	0 – 40	
Backfill φ'k	°	30	15 – 45	
Backfill γ	kN/m³	18	10 – …	
Backfill γsat	kN/m³	20	10 – …	
Surcharge on backfill (variable)	kN/m²	10	0 – …	e.g. 10 kN/m² for light traffic or storage.
Water level above base underside	m	0	0 – …	0 = drained backfill (no water pressure).
Foundation φ'k	°	32	10 – 45	
Foundation c'k	kN/m²	0	0 – …	
Foundation γ	kN/m³	19	10 – …	

Base friction factor k (tan δ = k tan φ')		1	0.3 – 1	1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)).
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	'National Annex' uses the approach of the annex chosen in the project details.
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4.
Displacement factor r		2.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped).
ψ2 for surcharge		0.3	0 – 1	
Concrete class		C30/37	C25/30 / C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Concrete unit weight	kN/m³	25	20 – …	
Cover — stem (earth face)	mm	50	20 – …	
Cover — base	mm	75	25 – …	
Stem bars Ø		16	7 …	
Stem bar spacing	mm	150	75 – 400	
Heel top bars Ø		16	7 …	
Heel bar spacing	mm	150	75 – 400	
Toe bottom bars Ø		12	7 …	
Toe bar spacing	mm	150	75 – 400	
Shear key bars Ø		12	7 …	
Shear key bar spacing	mm	200	75 – 400	

Checks performed (worked example)

Check	Clause	U (example)
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.73
DA2 eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.33
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.42

Overturning: M_dst,d ≤ M_stb,d	EN 1997-1 Eq. 2.4	0.39
Seismic sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.92
Seismic eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.4
Seismic bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.8
Seismic overturning: M_dst ≤ M_stb	EN 1998-5 7.3.1	0.37
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.67
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.38
Bending: M_Ed ≤ M_Rd	6.1	0.66
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.51
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.5
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.47
Bending: M_Ed ≤ M_Rd	6.1	0.49
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.44
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.39
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.84
Bending: M_Ed ≤ M_Rd	6.1	0.37
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.56
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6

Embedded pile wall with anchors

Standard: EN 1997-1 §9, EN 1992-1-1, EN 1998-5

Contiguous or secant bored pile wall, cantilever or with several anchor rows: elastic–plastic subgrade reaction analysis (displacements, earth pressures, moments, anchor forces), passive mobilisation, RC pile design and anchors. Interactive 3D model.

Available tools: 3D model · quantities and cost · automatic design (Embedment below excavation t, Pile diameter D)

Results that can be linked: H, t, D, a, EI, k_h, u_max, u_top, P_1,k, P_2,k, M_Ed, V_Ed, E_p,mob, E_p,d ...

Method and assumptions

- Beam on elastic–plastic Winkler springs: at-rest pressure ± k_h u bounded by active and passive; anchors as springs locked off at P_0.
- Factored effects (DA2: characteristic soil, effects × γ_G) or factored strength (DA1-C2, DA3).

Inputs

Input	Unit	Example	Range	Notes
Excavation depth H	m	7	1 – 25	
Embedment below excavation t	m	5	0.5 – …	
Pile diameter D	m	0.6	0.3 – 1.5	
Pile spacing a	m	0.75	0.3 – 3	Contiguous wall: a ≥ D (gap ≤ ~0.15 m in stiff soil); secant: a < D.
Anchor rows	table			Depth [m], Inclination α [°], Horizontal spacing [m], Lock-off load P_0 [kN], Free length [m], Tendon area A_t [mm²], Design resistance R_a,d [kN] — Leave the table empty for a cantilever wall. Strands Ø15.7: 150 mm² each.
Soil φ'k	°	32	15 – 45	
Soil c'k	kN/m²	5	0 – …	
Soil γ	kN/m³	19	10 – …	
Soil γ_sat	kN/m³	20	10 – …	
Groundwater depth behind the wall	m	20	0 – …	In front the water is lowered to the excavation level.
Surcharge behind the wall (variable)	kN/m²	10	0 – …	
Modulus of subgrade reaction k_h	MN/m³	25	2 – 200	Typical: loose sand 10, medium dense 25, dense 50–100, stiff clay 20–40 MN/m³.
Wall friction δ_a/φ'		0.667	0 – 0.667	
Wall friction δ_p/φ'		0.5	0 – 0.5	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Include seismic design (EN 1998-5)		no		
Design ground acceleration αgR·γI		0.2	0 – 0.6	
Soil factor S		1.2	1 – 1.8	

Concrete class		C30/37	C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Longitudinal bars		12	6 – 40	
Bar Ø		20	16 / 20 / 25 / 32	
Spiral Ø		10	8 / 10 / 12	
Spiral pitch	mm	150	50 – 400	
Cover to bar centre	mm	85	50 – 150	
Bending stiffness factor (cracking)		0.7	0.3 – 1	E_cm I × factor; 0.5–0.7 for cracked RC piles.

Checks performed (worked example)

Check	Clause	U (example)
DA2 embedment: E_p,mob ≤ E_p,d	EN 1997-1 9.7.4	0.38
DA2 anchor row 1: P_d ≤ R_a,d	EN 1997-1 8.5	0.6
DA2 anchor row 2: P_d ≤ R_a,d	EN 1997-1 8.5	0.73
Bending: M_Ed ≤ M_Rd	6.1	0.17
Shear: V_Ed ≤ max(V_Rd,c, V_Rd,s)	6.2	0.25

Foundation settlement

Standard: EN 1997-1 §6.6, Annex F, Annex H

Immediate and consolidation settlement of a rectangular footing or raft on layered soil: Boussinesq stress distribution, oedometer modulus or C_c/C_r/OCR per layer, influence depth and limits of EN 1997-1 Annex H. 3D stress bulb.

Available tools: 3D model

Results that can be linked: B, L, D, q, σ'_v0(D), q_net, z_inf, s_c, s

Method and assumptions

- Boussinesq stresses below the centre (Newmark), sub-layers of 0.25 m down to Δσ = 0.2 σ'_v0; granular by E_oed, clays by C_c/C_r/OCR.

Inputs

Input	Unit	Example	Range	Notes

Foundation width B	m	2.5	0.3 – …	
Foundation length L	m	2.5	0.3 – …	
Founding depth D	m	1	0 – …	
Contact pressure (SLS, quasi-permanent)	kN/m²	180	0 – …	Total pressure under the foundation for G + ψ2Q. The net pressure q − γD causes settlement.
Rigid foundation (s = 0.8 s_centre)		yes		A rigid footing settles uniformly ≈ 0.8 × the flexible centre value.
Groundwater depth below ground	m	4	0 – …	
Soil layers (from the ground surface down)	table			Thickness [m], Model, γ [kN/m³], E_oed (granular) [MPa], C_c (clay), C_r (clay), e_0, OCR
Limiting settlement	mm	50	5 – …	EN 1997-1 Annex H: up to 50 mm is often acceptable for normal structures on separate foundations; less for sensitive structures.

Checks performed (worked example)

Check	Clause	U (example)
Settlement: s ≤ s_lim	EN 1997-1 Annex H	0.34

Gabion wall

Standard: EN 1997-1, EN 1998-5, EN 10223-3

Gravity wall of stacked gabion courses with set-backs: Coulomb and Mononobe-Okabe thrust, sliding, overturning, bearing and the stability of every joint between courses. Interactive 3D model.

Available tools: 3D model · quantities and cost · DXF drawing

Results that can be linked: H, B, γ_s, n, γ_g, β, h_v, q_k, φ'_d, K_a, φ'_f,d, tan δ_d, H_d, R_h,d …

Method and assumptions

- Global stability as a gravity wall (γ_g = γ_s(1 − n)) plus sliding and eccentricity at every joint between courses.

Inputs

Input	Unit	Example	Range	Notes
Gabion courses (from the bottom up)	table			Width [m], Height [m], Front set-back from the course below [m] — Standard baskets are 0.5 m or 1.0 m high and 1.0–1.5 m wide modules.

Embedment of the base below ground in front	m	0.5	0 – …	
Wall length (3D view)	m	8	1 – …	
Backfill slope β	°	0	0 – 40	
Backfill φ'k	°	30	15 – 45	
Backfill γ	kN/m³	18	10 – …	
Backfill γsat	kN/m³	20	10 – …	
Surcharge on backfill (variable)	kN/m²	10	0 – …	e.g. 10 kN/m² for light traffic or storage.
Water level above base underside	m	0	0 – …	0 = drained backfill (no water pressure).
Foundation φ'k	°	32	10 – 45	
Foundation c'k	kN/m²	0	0 – …	
Foundation γ	kN/m³	19	10 – …	
Base friction factor k (tan δ = k tan φ')		1	0.3 – 1	1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)).
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	'National Annex' uses the approach of the annex chosen in the project details.
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4.
Displacement factor r		2.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped).
ψ2 for surcharge		0.3	0 – 1	
Unit weight of the stone	kN/m³	26	18 – 30	Limestone 24–26, basalt/granite 26–28 kN/m³.
Porosity of the fill n		0.35	0.25 – 0.45	0.30–0.40 for hand-placed, well-graded stone.
Friction angle between courses	°	35	20 – 45	Stone-on-stone contact through the mesh; 30–40°.

Checks performed (worked example)

Check	Clause	U (example)
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.64

DA2 eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.29
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.32
Overturning: M_dst,d ≤ M_stb,d	EN 1997-1 Eq. 2.4	0.31
Seismic sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.8
Seismic eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.33
Seismic bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.59
Seismic overturning: M_dst ≤ M_stb	EN 1998-5 7.3.1	0.29
Joint 1 sliding: H_Ed ≤ N_fav tan φ_j	EN 1997-1 6.5.3	0.42
Joint 1 eccentricity: |e| ≤ b_j/3	EN 1997-1 6.5.4	0.03
Joint 2 sliding: H_Ed ≤ N_fav tan φ_j	EN 1997-1 6.5.3	0.33
Joint 2 eccentricity: |e| ≤ b_j/3	EN 1997-1 6.5.4	0.05

Gravity retaining wall

Standard: EN 1997-1, EN 1998-5, EN 1992-1-1 §12

Mass concrete or masonry gravity wall with optional footing: Coulomb and Mononobe-Okabe thrust, groundwater, sliding, overturning, bearing and plain-concrete section checks. Interactive 3D model.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Width at base of wall body, Footing toe projection, Footing heel projection)

Results that can be linked: H, b_top, B_w, f, t_f, B, β, h_v, q_k, φ'_d, K_a, φ'_f,d, tan δ_d, H_d …

Method and assumptions

- Same stability engine as the cantilever wall; plain-concrete body sections to EN 1992-1-1 §12.

Inputs

Input	Unit	Example	Range	Notes
Wall height (above footing)	m	3	0.5 – 10	
Top width	m	0.5	0.2 – …	
Width at base of wall body	m	1.9	0.3 – …	
Front face batter (horizontal offset)	m	0.2	0 – …	0 = vertical front face.
Footing thickness (0 = no footing)	m	0.4	0 – …	
Footing toe projection	m	0.5	0 – …	

Footing heel projection	m	0.3	0 – …	
Soil depth over toe	m	0.5	0 – …	
Wall length (3D view)	m	8	1 – …	
Backfill slope β	°	0	0 – 40	
Backfill φ'k	°	30	15 – 45	
Backfill γ	kN/m³	18	10 – …	
Backfill γsat	kN/m³	20	10 – …	
Surcharge on backfill (variable)	kN/m²	10	0 – …	e.g. 10 kN/m² for light traffic or storage.
Water level above base underside	m	0	0 – …	0 = drained backfill (no water pressure).
Foundation φ'k	°	32	10 – 45	
Foundation c'k	kN/m²	0	0 – …	
Foundation γ	kN/m³	19	10 – …	
Base friction factor k (tan δ = k tan φ')		1	0.3 – 1	1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)).
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	'National Annex' uses the approach of the annex chosen in the project details.
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4.
Displacement factor r		2.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped).
ψ2 for surcharge		0.3	0 – 1	
Concrete class		C20/25	C12/15 / C16/20 / C20/25 / C25/30 / C30/37	
Unit weight of wall	kN/m³	24	15 – …	24 plain concrete, 22 stone masonry, 20 gabions (approx.).

Checks performed (worked example)

Check	Clause	U (example)
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.68

DA2 eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.37
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.42
Overturning: M_dst,d ≤ M_stb,d	EN 1997-1 Eq. 2.4	0.39
Seismic sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.83
Seismic eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.42
Seismic bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.72
Seismic overturning: M_dst ≤ M_stb	EN 1998-5 7.3.1	0.36
Body at base of wall body: N_Ed ≤ N_Rd	12.6.1	0.01
Body at base of wall body: |e| ≤ 0.3 b_w	12.6.5.1	0.64
Body at base of wall body: H_Ed ≤ N_Ed tan 35° (joint)	12.6.3	0.46
Body at mid-height: N_Ed ≤ N_Rd	12.6.1	0.01
Body at mid-height: |e| ≤ 0.3 b_w	12.6.5.1	0.55
Body at mid-height: H_Ed ≤ N_Ed tan 35° (joint)	12.6.3	0.45

Liquefaction potential

Standard: EN 1998-5 §4.1.4, Annex B

Saturated sandy layers from SPT blow counts: overburden and fines corrections, cyclic resistance with the magnitude factor C_M, seismic shear stress τ_e and the factor of safety against liquefaction for each layer. Interactive 3D soil column.

Available tools: 3D model

Results that can be linked: a_gR, α, S, αS, C_M, λ

Method and assumptions

- τ_e = 0.65 α S σ_v0; (N1)60 with C_N and fines correction; CRR of Youd et al. (2001) × C_M (Table B.1); FS ≥ λ of the National Annex.

Inputs

Input	Unit	Example	Range	Notes
Reference ground acceleration a_gR		0.25	0 – 0.6	In g; zone buttons of the National Annex.
Importance class		II	I / II / III / IV	
Soil factor S		1.15	1 – 1.8	EN 1998-1 Table 3.2 (ground type C: 1.15).
Surface wave magnitude M_s		6.5	5.5 / 6.0 / 6.5 / 7.0 / 7.5 / 8.0	Design earthquake magnitude for the site.
Groundwater depth	m	2	0 – …	

Soil layers (from the surface down)	table			Thickness [m], Soil, γ [kN/m³], SPT N_60, Fines content [%] — N_60: SPT blow count corrected to 60 % energy. Clays (FC > 35 % plastic) are not checked.

Checks performed (worked example)

Check	Clause	U (example)
Layer 2 at z = 4.0 m: FS ≥ λ	EN 1998-5 4.1.4(11)	0.64
Layer 3 at z = 7.5 m: FS ≥ λ	EN 1998-5 4.1.4(11)	0.79
Layer 4 at z = 12.0 m: FS ≥ λ	EN 1998-5 4.1.4(11)	0.86

Pad footing

Standard: EN 1997-1, EN 1992-1-1, EN 1998-5

Isolated reinforced concrete footing under a column with biaxial moments: effective-area bearing (Annex D, shape and inclination factors), sliding, eccentricity, bending, beam shear and punching. Interactive 3D model.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Footing length Bx, Footing width By, Footing thickness h)

Results that can be linked: B_x, B_y, h, D, c_x, c_y, W_f, W_s, φ'_d, M_x,d, M_y,d, H_d, e_x, e_y …

Method and assumptions

- Effective area B'×L' for biaxial eccentricity (Annex D); punching with the critical perimeter search between 0 and 2d (6.4.4).

Inputs

Input	Unit	Example	Range	Notes
Footing length Bx	m	2.4	0.5 – …	
Footing width By	m	2.4	0.5 – …	
Footing thickness h	m	0.6	0.25 – …	
Founding depth D (ground to underside)	m	1.2	0.3 – …	
Column size cx	m	0.4	0.15 – …	
Column size cy	m	0.4	0.15 – …	
Axial load — permanent G	kN	700	0 – …	Loads at the top of the footing (column base), downward positive.
Axial load — variable Q	kN	300	0 – …	

Moment Mx — permanent	kNm	30		Mx causes eccentricity along X (rotation about the Y axis).
Moment Mx — variable	kNm	20		
Moment My — permanent	kNm	0		My causes eccentricity along Y.
Moment My — variable	kNm	10		
Horizontal load Hx — permanent	kN	10		
Horizontal load Hx — variable	kN	10		
Include seismic combination		yes		
Axial load N_E	kN	750	0 – …	Design values from the seismic analysis (G + ψ2Q + E), capacity design as required by EN 1998-1 4.4.2.6.
Moment Mx,E	kNm	150		
Moment My,E	kNm	50		
Horizontal Hx,E	kN	80		
Soil φ'k	°	30	10 – 45	
Soil c'k	kN/m²	5	0 – …	
Soil γ	kN/m³	19	10 – …	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover (bottom)	mm	50	25 – …	
Bars in X Ø		14	10 / 12 / 14 / 16 / 20 / 25	
Spacing in X	mm	150	75 – 400	
Bars in Y Ø		14	10 / 12 / 14 / 16 / 20 / 25	
Spacing in Y	mm	150	75 – 400	

Checks performed (worked example)

Check	Clause	U (example)
DA2 eccentricity: $(e_x/B_x)^2 + (e_y/B_y)^2 \le 1/9$	EN 1997-1 6.5.4	0
DA2 bearing: $q_{Ed} \le q_{Rd}$	EN 1997-1 6.5.2	0.36
DA2 sliding: $H_d \le R_{h,d}$	EN 1997-1 6.5.3	0.06
Seismic eccentricity: $(e_x/B_x)^2 + (e_y/B_y)^2 \le 1/9$	EN 1997-1 6.5.4	0.08
Seismic bearing: $q_{Ed} \le q_{Rd}$	EN 1997-1 6.5.2	0.42
Seismic sliding: $H_d \le R_{h,d}$	EN 1997-1 6.5.3	0.19
Reinforcement: $A_{s,prov} \ge A_{s,req}$	6.1	0.6
Minimum reinforcement: $A_{s,prov} \ge A_{s,min}$	9.2.1.1	0.71
Bending: $M_{Ed} \le M_{Rd}$	6.1	0.58
Shear: $V_{Ed} \le V_{Rd,c}$	6.2.2	0.64
Reinforcement: $A_{s,prov} \ge A_{s,req}$	6.1	0.55
Minimum reinforcement: $A_{s,prov} \ge A_{s,min}$	9.2.1.1	0.69
Bending: $M_{Ed} \le M_{Rd}$	6.1	0.54
Shear: $V_{Ed} \le V_{Rd,c}$	6.2.2	0.61
Punching: $v_{Ed} \le v_{Rd}$	6.4.4(2)	0.58
Punching at column face: $v_{Ed,0} \le v_{Rd,max}$	6.4.5(3)	0.41

Raft foundation on springs

Standard: EN 1997-1 §6, EN 1992-1-1

Raft under a regular column grid on a Winkler subgrade: strip analysis in both directions (settlement, soil pressure, moments), bottom and top reinforcement, punching at the columns and bearing pressure. 3D settlement map.

Available tools: 3D model · quantities and cost

Results that can be linked: $L_x \times L_y$, t, k_s, q_{avg}, s_{max}, q_{Ed}, m_{Ed}, μ, μ_{lim}, ξ, z, $A_{s,req}$, $A_{s,min}$, $A_{s,max}$ …

Method and assumptions

- Strip method: each column line as a beam on Winkler springs carrying the full column loads in both directions (conservative); $E_c = 0.5\ E_{cm}$.

Inputs

Input	Unit	Example	Range	Notes
Columns in x		4	2 – 10	

Columns in y		3	2 – 10	
Column spacing in x	m	5	2 – 12	
Column spacing in y	m	5	2 – 12	
Overhang beyond the outer columns	m	0.8	0 – 3	
Raft thickness	mm	700	250 – 2500	
Column size (square)	mm	450	200 – …	
Design column load N_Ed (interior)	kN	2200	0 – …	Edge columns take 60 % and corner columns 35 % of this load.
Characteristic (SLS) column load	kN	1550	0 – …	
Modulus of subgrade reaction k_s	MN/m³	15	1 – 300	Soft clay 5–15, medium 15–40, dense sand 40–100 MN/m³.
Design bearing resistance q_Rd	kN/m²	300	20 – …	From a bearing capacity calculation (e.g. the strip footing module).
Limiting settlement	mm	50	5 – …	
Concrete class		C30/37	C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover	mm	50	30 – …	
Bottom bars Ø (both ways)		16	12 / 14 / 16 / 20 / 25	
Bottom bar spacing	mm	150	75 – 300	
Top bars Ø (both ways)		16	12 / 14 / 16 / 20 / 25	
Top bar spacing	mm	150	75 – 300	

Checks performed (worked example)

Check	Clause	U (example)
Settlement s_max ≤ s_lim	EN 1997-1 Annex H	0.1
Bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.35
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.52
Minimum reinforcement	9.3.1.1	0.7
Bending: m_Ed ≤ m_Rd	6.1	0.51
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.44
Minimum reinforcement	9.3.1.1	0.7
Bending: m_Ed ≤ m_Rd	6.1	0.43
Punching: v_Ed ≤ v_Rd	6.4.4(2)	0.75

Punching at column face: v_Ed,0 ≤ v_Rd,max	6.4.5(3)	0.37

Single pile — axial and lateral

Standard: EN 1997-1 §7, EN 1992-1-1

Bored, CFA or driven pile in layered clay/sand: shaft and base resistance from soil parameters (α-method, β-method, Meyerhof N_q*), partial and correlation/model factors of the National Annex, tension, lateral resistance (Broms) and structural check. Interactive 3D model.

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Pile diameter D, Pile length L (below ground))

Results that can be linked: D, L, z_w, R_s,cal, q_b, R_b,cal, W_p, ΔW, γ_Rd, R_b,k, R_s,k, F_c,d, R_c,d, M_y …

Method and assumptions

- Shaft: α-method (clay) and β-method K tan δ σ'_v (sand); base: 9c_u or Meyerhof N_q* with base efficiency η_b (0.5 bored/CFA, 1.0 driven).
- Characteristic value by model factor (7.6.2.3(8)) or ξ3/ξ4; lateral by Broms (free head, top layer).

Inputs

Input	Unit	Example	Range	Notes
Pile type		Bored	Bored / CFA / Driven	
Pile diameter D	m	0.8	0.2 – 3	
Pile length L (below ground)	m	18	2 – 80	
Concrete class		C30/37	C20/25 / C25/30 / C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Longitudinal bars		10	6 – 40	
Bar Ø		20	12 / 14 / 16 / 20 / 25	
Cover to bar centre	mm	75	40 – 150	Bored piles: cover 75 mm to the links is usual (EN 1536).
Axial load — permanent G	kN	800	0 – …	

Axial load — variable Q	kN	250	0 – …	
Uplift (tension) load — variable	kN	0	0 – …	e.g. wind uplift; checked against shaft resistance in tension.
Horizontal load — permanent	kN	20	0 – …	
Horizontal load — variable	kN	30	0 – …	
Height of the horizontal load above ground e	m	0	0 – …	
Groundwater depth below ground	m	3	0 – …	
Soil layers (from the ground surface down)	table			Thickness [m], Soil, γ [kN/m³], φ' (sand) [°], c_u (clay) [kN/m²]
Adhesion factor α (clay)		0.5	0.2 – 1	Typical 0.45–0.5 for bored piles in stiff clay; 0.5–1.0 driven.
Characteristic resistance		Alternative method (model factor)	Alternative method (model factor) / Ground test profiles (ξ3, ξ4)	Alternative method: R_k = R_cal / γ_Rd (7.6.2.3(8)). Ground test profiles: R_k = min(R_mean/ξ3, R_min/ξ4) (7.6.2.3(5)).
Model factor γ_Rd		1.4	1 – 2	Cyprus NA 2.19: at least 1.25. About 1.4 when no load tests are made.
Number of test profiles n		1	1 – 10	
Stiff pile cap (ξ / 1.1, 7.6.2.3(7))		no		
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	

Checks performed (worked example)

Check	Clause	U (example)
DA2 compression: F_c,d ≤ R_c,d	7.6.2.1	0.8
DA2 lateral: H_d ≤ H_Rd	EN 1997-1 7.7	0.29
Minimum reinforcement A_s ≥ 0.5 % A_c (A_c ≤ 0.5 m²)	EN 1992-1-1 9.8.5 Table 9.6N	0.76

Axial: N_Ed ≤ N_Rd	6.1	0.14
Bending with axial force: M_Ed ≤ M_Rd(N_Ed)	6.1	0.17

Slope stability (Bishop)

Standard: EN 1997-1 §11, EN 1998-5 §4.1.3

Circular slip surfaces by Bishop's simplified method with automatic search of the critical circle; pore pressure ratio, crest surcharge and seismic pseudo-static action. Interactive 3D model.

Available tools: 3D model

Results that can be linked: H, β, L_s, φ'_k, c'_k, γ, r_u, q_k, F_k, F', F_d, k_h, k_v, F_d,E

Method and assumptions

- Bishop simplified method with automatic search of the critical circle; pseudo-static seismic action (EN 1998-5 4.1.3.3); optional soil nails / anchors adding T·r about the centre.

Inputs

Input	Unit	Example	Range	Notes
Slope height	m	6	1 – 50	
Slope angle	°	30	5 – 75	Angle of the slope face to the horizontal (1:√3 ≈ 30°, 1:2 ≈ 26.6°).
Depth to firm stratum below toe	m	6	0 – …	Slip circles are not allowed below this level.
Slope length (3D view)	m	20	4 – …	
Soil φ'k	°	30	0 – 45	
Soil c'k	kN/m²	8	0 – …	
Soil γ	kN/m³	19	10 – …	
Pore pressure ratio r_u		0	0 – 0.6	u = r_u·γ·h. 0 = dry/drained, 0.2–0.3 typical with a high water table.
Crest surcharge (variable)	kN/m²	10	0 – …	
Distance of surcharge from crest	m	1	0 – …	
Surcharge width	m	5	0 – …	
Soil nails / anchors	table			Head height above the toe [m], Inclination below horizontal [°], Length [m], Horizontal spacing [m], Design tensile

(optional)				resistance T_d [kN], Design pull-out resistance per metre [kN/m] — Leave empty for an unreinforced slope. Each row adds T = min(T_d, bond × length behind the slip surface) with its lever arm about the circle centre.
Include seismic (pseudo-static)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	
Topography factor S_T		1	1 – 1.4	EN 1998-5 Annex A: 1.0 for slopes < 15°, up to 1.2–1.4 for high ridges.

Checks performed (worked example)

Check	Clause	U (example)
Persistent: E_d ≤ R_d (1 ≤ F_d)	EN 1997-1 Eq. 2.5	0.8
Seismic: E_d ≤ R_d (1 ≤ F_d,E)	EN 1998-5 4.1.3.3	0.81

Strip footing and tie beams

Standard: EN 1997-1, EN 1992-1-1, EN 1998-5

Continuous footing under a wall (per metre run): bearing with load inclination, sliding, eccentricity, transverse bending and shear. Optional seismic tie beams between footings (EN 1998-5 5.4.1.2).

Available tools: 3D model · quantities and cost · DXF drawing · automatic design (Footing width B, Footing thickness h)

Results that can be linked: B, h, D, b_w, W_f, W_s, φ'_d, M_d, H_d, e, B', N_q, i_q, q_Rd …

Method and assumptions

- Strip bearing per metre run with load inclination; tie-beam force ±k α S N (EN 1998-5 5.4.1.2) with the minimum sections of EN 1998-1 5.8.2.

Inputs

Input	Unit	Example	Range	Notes
Footing width B	m	1.4	0.4 – …	
Footing thickness h	m	0.45	0.2 – …	
Founding depth D (ground to underside)	m	1	0.3 – …	
Wall thickness b_w	m	0.25	0.1 – …	

Line load — permanent G	kN/m	180	0 – …	At the top of the footing (base of the wall), downward positive.
Line load — variable Q	kN/m	60	0 – …	
Moment — permanent	kNm/m	5		Moment about the footing axis (across the width B).
Moment — variable	kNm/m	5		
Horizontal load — permanent	kN/m	0		
Horizontal load — variable	kN/m	5		
Include seismic combination		no		
Line load N_E	kN/m	200	0 – …	G + ψ2Q + E from the seismic analysis.
Moment M_E	kNm/m	30		
Horizontal H_E	kN/m	25		
Soil φ'k	°	30	10 – 45	
Soil c'k	kN/m²	5	0 – …	
Soil γ	kN/m³	19	10 – …	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover (bottom)	mm	50	25 – …	
Transverse bars Ø		12	10 / 12 / 14 / 16 / 20 / 25	
Spacing of transverse bars	mm	200	75 – 400	
Design tie beams (seismic)		yes		Tie beams connect the footings so they move together in an earthquake.
Reference ground acceleration a_gR		0.25	0 – 0.6	In g. Use the zone buttons.
Ground type		C	B / C / D	Ground type A: tie beams may be omitted (EN 1998-5 5.4.1.2(5)).
Importance class		II	I / II / III / IV	
Design axial load of column/wall 1	kN	900	0 – …	Seismic design situation, the two elements the beam joins.

Design axial load of column/wall 2	kN	700	0 – …	
Tie beam width	m	0.3	0.15 – …	
Tie beam depth	m	0.5	0.2 – …	
Storeys above the basement		3	1 – 40	
Bars top + bottom (total)		8	4 – 16	Half at the top, half at the bottom.
Tie beam bars Ø		16	10 / 12 / 14 / 16 / 20 / 25	

Checks performed (worked example)

Check	Clause	U (example)
DA2 eccentricity: e/B ≤ 1/3	EN 1997-1 6.5.4	0.1
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.56
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.07
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.5
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.93
Bending: M_Ed ≤ M_Rd	6.1	0.48
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.33
Tie beam width b ≥ 0.25 m	EN 1998-1 5.8.2(4)	0.83
Tie beam depth h ≥ 0.4 m (3 storeys)	EN 1998-1 5.8.2(4)	0.8
Minimum ratio top and bottom ρ_b ≥ 0.4 %	EN 1998-1 5.8.2(5)	0.75
Tie beam in tension: N_tie ≤ N_t,Rd	EN 1992-1-1 6.1	0.13
Tie beam in compression: N_tie ≤ N_c,Rd	EN 1992-1-1 6.1	0.03

LOADS

Floor and roof loads

Standard: EN 1991-1-1

Self-weight of the floor build-up from the material densities of Annex A, imposed load by category of use (Tables 6.1–6.10), reduction factors α_A and α_n, movable partitions, parapet loads, and the design values.

Available tools: 3D model

Results that can be linked: g_layers, g_k, q_k, Q_k, A, α_A, q_k,part, q_k,tot, p_Ed, p_k, p_qp, q_k,par

Method and assumptions

- Self-weight $g_k = \Sigma\ \gamma\ t$ of the build-up with the nominal densities of EN 1991-1-1 Annex A (upper value of a range); imposed q_k, Q_k by category (Tables 6.2, 6.4, 6.8, 6.10, National Annex value or the recommended one).
- $\alpha_A = 5/7\ \psi_0 + A_0/A \le 1.0$ (≥ 0.6 for C, D; Eq. 6.1) for floors, $\alpha_n = (2 + (n - 2)\ \psi_0)/n$ for columns (Eq. 6.2); movable partitions 0.5 / 0.8 / 1.2 kN/m² (6.3.1.2(8)) added unreduced; parapets Table 6.12, car-park barriers Annex B.

Inputs

Input	Unit	Example	Range	Notes
Category of use		A — residential: floors	15 …	EN 1991-1-1 Tables 6.1, 6.3, 6.7 and 6.9. Areas with several uses: take the most unfavourable category (6.3.1.2(7)).
Values of q_k and Q_k		Recommended values (National Annex)	Recommended values (National Annex) / My own values	Where Table 6.2 gives a range the National Annex chooses the value; the recommended (underlined) value is used otherwise.
Imposed load q_k	kN/m²	2	0 – …	
Concentrated load Q_k	kN	2	0 – …	
Movable partitions		Movable partitions ≤ 2.0 kN/m	None / Movable partitions ≤ 1.0 kN/m / Movable partitions ≤ 2.0 kN/m / Movable partitions ≤ 3.0 kN/m / Heavier partitions (> 3.0 kN/m)	Self-weight per metre length of wall; taken as an extra uniform imposed load (6.3.1.2(8)). Heavier partitions must be modelled where they stand.
Load for		Floor, beam or slab	Floor, beam or slab / Column or wall (several storeys)	
Loaded area A of the member	m²	20	0.1 – …	Area that loads the member under consideration (α_A, Eq. 6.1).
Storeys of the same		3	1 – 60	α_n reduces the imposed load from

category above the member				several storeys (Eq. 6.2, n > 2).
Apply the reduction factor		yes		When the imposed load is an accompanying action only one of ψ and α_n may be used (3.3.2(2)P).
Floor build-up (permanent)	table			Layer, Material (EN 1991-1-1 Annex A), Thickness [mm], γ (only for 'Other material') [kN/m³] — Paste from Excel or add rows. Ranges in Annex A: the upper value is used.
Other permanent loads (ceiling, services)	kN/m²	0	0 – …	
Horizontal load on parapets / partitions		yes		Table 6.12 (height ≤ 1.20 m) and Annex B for car parks.

Load combinations

Standard: EN 1990 Annex A1

Generates all ULS (6.10), seismic (6.12b) and SLS (characteristic, frequent, quasi-permanent) combinations from your actions, with the ψ factors of the National Annex, and finds the governing values.

Available tools: 3D model

Method and assumptions

- All ULS (6.10), seismic (6.12b) and SLS combinations with ψ of the annex; each variable action in turn as leading action.

Inputs

Input	Unit	Example	Range	Notes
Actions (characteristic values, one effect e.g. N, M or V)	table			Name, Type, Category (imposed), Value

Site altitude (for snow ψ)		100	0 – …	m
Unit of the values		kN	kN / kNm / kN/m / kN/m²	

Wind and snow loads at the site

Standard: EN 1991-1-4, EN 1991-1-3

Peak velocity pressure q_p(z), wall pressures and roof snow load. With the Cyprus National Annex v_b,0 and s_k are taken automatically from the project site on the official maps.

Available tools: 3D model

Results that can be linked: v_b,0, v_b, z_0, k_r, c_r(z), v_m(z), I_v(z), q_p(z), c_e(z), h/d, c_pe,10 (D), c_pe,10 (E), w_D, w_E …

Method and assumptions

- q_p(z) by Eq. 4.8 with c_dir = c_season = 1; walls by Table 7.1; roof snow μ_1 C_e C_t s_k.

Inputs

Input	Unit	Example	Range	Notes
Use the project site (maps of the National Annex)		yes		Pick the site in the project details (📍 Map). Otherwise give v_b,0 and s_k.
Fundamental basic wind velocity v_b,0		24	10 – 60	m/s — used when no site is selected or for EN values.
Site altitude above sea level		100	0 – 3000	m
Ground snow load s_k (manual)	kN/m²	0.5	0 – …	Used only when the National Annex gives no snow map/formula.
Terrain category		III	0 / I / II / III / IV	0 sea, I lakes/flat, II open country, III suburbs/villages, IV cities (≥15 % covered by buildings > 15 m).
Orography factor c_o		1	1 – 2	1.0 unless the site is on a hill or cliff (Annex A.3).
Reference height z (building height)	m	9	1 – 200	
Building depth in the wind direction d	m	12	1 – …	
Roof pitch α	°	0	0 – 80	

Exposure (snow)		Normal	Windswept / Normal / Sheltered	

MASONRY

Masonry wall

Standard: EN 1996-1-1

Load-bearing unreinforced masonry wall: f_k from units and mortar, γ_M by category and class, effective height and thickness, slenderness, reduction factors at top/bottom/mid-height (Annex G) and in-plane shear of a shear wall. Interactive 3D model.

Available tools: 3D model · quantities and cost · automatic design (Wall thickness t)

Results that can be linked: K, f_k, γ_M, f_d, ρ_n, h_ef, t_ef, h_ef / t_ef, N_m,d, e_init, e_i,top, e_i,bot, Φ_top, e_k …

Method and assumptions

- Vertical resistance with the reduction factors Φ at top, bottom and mid-height (Annex G, E = K_E f_k); in-plane shear by Eq. 6.13.

Inputs

Input	Unit	Example	Range	Notes
Masonry unit		Clay	Clay / Calcium silicate / Aggregate concrete / Autoclaved aerated concrete / Manufactured stone / Dimensioned natural stone	
Unit group		Group 2	Group 1 / Group 2 / Group 3 / Group 4	Group 1 solid (voids ≤ 25 %), 2 vertically perforated ≤ 55 %, 3 ≤ 70 %, 4 horizontally perforated.
Normalised unit strength f_b	N/mm²	10	2 – 60	
Mortar class		M10	M2.5 / M5 / M10 / M15 / M20	
Units and mortar		Category I units, designed mortar	Category I units, designed mortar / Category I units,	

			prescribed mortar / Category II units, any mortar	
Execution control class		2	1 / 2 / 3 / 4 / 5	1 = best supervision … 5 = least.
Final creep coefficient φ∞		1	0.3 – 3	Clay 0.5–1.5, concrete/calcium silicate 1.0–2.0 (Table 6 CYS).
Wall thickness t	mm	250	90 – 600	
Clear storey height h	m	3	1 – 8	
Wall length l_w	m	4	0.3 – …	
Restraint of the wall		Top and bottom (RC floors)	Top and bottom (RC floors) / Top and bottom (timber floors) / Top, bottom and one vertical edge / Top, bottom and both vertical edges	
Design vertical load at the top N_Ed	kN/m	180	0 – …	
Eccentricity of the floor load at the top	mm	15	0 – …	From the floor bearing (e.g. t/6 for a slab bearing on part of the wall).
In-plane shear on the wall V_Ed	kN	60	0 – …	0 = not a shear wall.
Unit weight of masonry	kN/m³	18	5 – 25	

Checks performed (worked example)

Check	Clause	U (example)
Slenderness: h_ef/t_ef ≤ 27	5.5.1.4	0.33
Vertical load at top: N_Ed ≤ Φ t f_d	Eq. 6.2	0.32
Vertical load at mid-height: N_Ed ≤ Φ t f_d	Eq. 6.2	0.34
Vertical load at bottom: N_Ed ≤ Φ t f_d	Eq. 6.2	0.33

In-plane shear: V_Ed ≤ V_Rd	6.2	0.17

STEEL

Bolted connection

Standard: EN 1993-1-8 §3

Bolt group in single or double shear with eccentric shear and tension: spacing rules, bolt shear, bearing, tension, punching, combined shear and tension, long joints, block tearing and the net section of the plate. Interactive 3D model with bolt forces.

Available tools: 3D model

Results that can be linked: d_0, f_ub, f_u, V_Ed / n, F_v,Ed, F_v,Rd, F_b,Rd, b, N_pl,Rd, N_u,Rd, V_eff,1,Rd

Method and assumptions

- Elastic distribution of eccentric shear in the bolt group; bearing checked for the governing bolt with the end/inner values of Table 3.4.

Inputs

Input	Unit	Example	Range	Notes
Bolt class		8.8	4.6 / 5.6 / 8.8 / 10.9	
Bolt diameter		M20	8 …	
Threads in the shear plane		yes		
Shear planes		1	1 / 2	2 = double cover plates (splice).
Bolt rows in the force direction n1		3	1 – 12	
Bolt columns across n2		2	1 – 8	
Pitch p1 (in force direction)	mm	70	20 – …	
Pitch p2 (across)	mm	70	20 – …	
End distance e1	mm	50	10 – …	
Edge distance e2	mm	40	10 – …	
Thinnest plate thickness t	mm	12	4 – 60	
Plate steel grade		S275	S235 / S275 / S355	
Shear force V_Ed (along the force direction)	kN	300	0 – …	
Eccentricity of V from the bolt-group centroid	mm	0	0 – …	Causes a torsional moment on the group (elastic method).

Tension per bolt F_t,Ed	kN	0	0 – …	

Checks performed (worked example)

Check	Clause	U (example)
e_1 ≥ 1.2 d_0	Table 3.3	0.53
e_2 ≥ 1.2 d_0	Table 3.3	0.66
p_1 ≥ 2.2 d_0	Table 3.3	0.69
p_2 ≥ 2.4 d_0	Table 3.3	0.75
p_1, p_2 ≤ min(14t, 200)	Table 3.3	0.42
Bolt shear: F_v,Ed ≤ F_v,Rd	Table 3.4	0.53
Bearing: F_v,Ed ≤ F_b,Rd	Table 3.4	0.32
Group: V_Ed ≤ n min(F_v,Rd, F_b,Rd)	3.7	0.53
Plate tension: V_Ed ≤ min(N_pl,Rd, N_u,Rd)	EN 1993-1-1 6.2.3	0.76
Block tearing: V_Ed ≤ V_eff,1,Rd	3.10.2	0.42

Column base plate

Standard: EN 1993-1-8 §6.2.5–6.2.8, EN 1992-1-1 §6.7

Base plate of an I/H column on a concrete foundation under axial force, strong-axis moment and shear: bearing strength f_jd, effective width c, compression T-stubs, tension T-stubs with anchor bolts, moment resistance and shear by friction and bolts. Interactive 3D model.

Available tools: 3D model · quantities and cost · automatic design (Plate thickness t_p)

Results that can be linked: k_j, f_jd, c, F_C,Rd, N_j,Rd, m_x, l_eff,1, M_pl,1,Rd, F_T,1-2,Rd, F_T,3,Rd, F_T,Rd, z, F_t,Ed, F_c,Ed …

Method and assumptions

- Component method: compression T-stubs on the foundation, tension T-stub with anchor bolts (no prying), M–N from statics about the compressed flange.
- Anchorage of the bolts in concrete (EN 1992-4) and welds are not checked.

Inputs

Input	Unit	Example	Range	Notes
Column section		HEB 240	40 …	
Axial compression N_Ed	kN	600	0 – …	
Moment M_y,Ed	kNm	100	0 – …	
Shear V_Ed	kN	40	0 – …	
Plate width b_p (along the flanges)	mm	400	150 – …	

Plate length l_p (along the web)	mm	500	150 – …	
Plate thickness t_p	mm	25	8 – 80	
Plate steel grade		S275	S235 / S275 / S355	
Anchor bolt class		8.8	4.6 / 5.6 / 8.8 / 10.9	
Anchor bolt diameter		M24	7 …	
Bolts per side (in each tension row)		2	1 – 4	
Bolt distance from plate end e_x	mm	50	25 – …	
Bolt gauge across w	mm	240	60 – …	
Foundation concrete		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Foundation width	m	1.2	0.3 – …	
Foundation length	m	1.2	0.3 – …	
Flange weld throat a	mm	8	3 – 20	

Checks performed (worked example)

Check	Clause	U (example)
Pure compression: N_Ed ≤ N_j,Rd	6.2.8.2	0.28
Tension side: F_t,Ed ≤ F_T,Rd	Table 6.7	0.44
Compression side: F_c,Ed ≤ F_C,Rd	Table 6.7	0.69
Shear: V_Ed ≤ F_f,Rd + F_vb,Rd	6.2.2(8)	0.12

Fillet weld group

Standard: EN 1993-1-8 §4.5

Rectangular group of fillet welds under axial force, shear and bending: throat stresses σ⊥, τ⊥, τ∥ by the directional method (Eq. 4.1) and the simplified method, with the weld detailing rules. Interactive 3D model.

Available tools: 3D model · quantities and cost

Results that can be linked: A_w, I_w, f_u, β_w, σ_w, σ⊥ = τ⊥, τ∥, σ_eq, f_vw,d, F_w,Ed / a, utilisation

Method and assumptions

- Elastic stresses on the throat area unfolded into the joint plane; σ⊥ = τ⊥ = σ_w/√2, τ∥ from the shear on the vertical welds; Eq. 4.1 and the simplified method 4.5.3.3.

Inputs

Input	Unit	Example	Range	Notes

Width of the weld group b (horizontal welds)	mm	200	20 – …	
Height of the weld group h (vertical welds)	mm	300	20 – …	
Weld along the top (horizontal)		yes		
Weld along the bottom (horizontal)		yes		
Welds along both sides (vertical)		yes		
Throat thickness a	mm	5	3 – 20	a ≥ 3 mm (4.5.2(2)); leg length ≈ 1.41 a.
Steel grade (weaker part)		S275	S235 / S275 / S355	
Thickness of the thinner part	mm	12	3 – …	
Axial force N_Ed (tension +)	kN	150		
Shear force V_Ed (vertical)	kN	120	0 – …	
Moment M_Ed (about the horizontal axis)	kNm	20		

Checks performed (worked example)

Check	Clause	U (example)
Throat a ≥ 3 mm	4.5.2(2)	0.6
Throat a ≤ 0.7 t (practical maximum)	good practice	0.6
Weld length b ≥ max(30 mm, 6a)	4.5.4(1)	0.15
Weld length h ≥ max(30 mm, 6a)	4.5.4(1)	0.1
Weld: $\sqrt{(\sigma_\perp^2 + 3(\tau_\perp^2 + \tau_\parallel^2))} \le f_u/(\beta_w \gamma_{M2})$	Eq. 4.1	0.31
Weld: $\sigma_\perp \le 0.9 f_u / \gamma_{M2}$	Eq. 4.1	0.17

Steel beam design

Standard: EN 1993-1-1:2005

Simply supported rolled I-beam: classification, bending, shear, bending-shear interaction, lateral-torsional buckling and deflection.

Available tools: 3D model · quantities and cost · automatic design (Section)

Results that can be linked: f_y, ε, E, G, c_f/t_f, c_w/t_w, W_y, g_sw, g_k, q_k, w_Ed, M_Ed, V_Ed, M_c,Rd …

Method and assumptions

- Simply supported rolled I-beam; LTB with M_cr (NCCI SN003) and 6.3.2.3.

Inputs

Input	Unit	Example	Range	Notes
Section		IPE 300	40 …	
Steel grade		S275	S235 / S275 / S355 / S460	
Span	m	6	0.1 – …	
LTB effective length (0 = unrestrained span)	m	2	0 – …	Distance between lateral restraints to the compression flange.
Permanent UDL (excl. self weight)	kN/m	8	0 – …	
Variable UDL	kN/m	10	0 – …	
Permanent point load at midspan	kN	0	0 – …	
Variable point load at midspan	kN	0	0 – …	
γG		1.35	0 – …	
γQ		1.5	0 – …	
γM0		1	0.5 – …	
γM1		1	0.5 – …	
Variable load deflection limit L/…		360	1 – …	

Checks performed (worked example)

Check	Clause	U (example)
Bending: M_Ed ≤ M_c,Rd	6.2.5(1)	0.69
Shear: V_Ed ≤ V_pl,Rd	6.2.6(1)	0.19
Shear buckling: h_w/t_w ≤ 72ε/η	6.2.6(6)	0.59
LTB: M_Ed ≤ M_b,Rd	6.3.2.1(1)	0.73
Deflection: δ_q ≤ δ_lim	EN 1990 A1.4	0.58

Steel column

Standard: EN 1993-1-1 §6.3, Annex B

Rolled I/H column under axial force and biaxial end moments: classification, cross-section resistance, flexural buckling about both axes, lateral-torsional buckling and the interaction of Eq. 6.61/6.62 (Annex B). Interactive 3D model with buckled shape.

Available tools: 3D model · quantities and cost · automatic design (Section)

Results that can be linked: f_y, ε, c/t (flange), c/t (web), N_c,Rd, M_y,Rd, M_z,Rd, V_pl,z,Rd, L_cr,y, λ̄_y, χ_y, N_b,y,Rd, L_cr,z, λ̄_z …

Method and assumptions

- Buckling curves of Table 6.2; interaction factors of Annex B (Method 2).
- Class 4 webs by the effective width of EN 1993-1-5.

Inputs

Input	Unit	Example	Range	Notes
Section		HEB 240	40 …	
Steel grade		S355	S235 / S275 / S355 / S460	
Column height L	m	4	0.5 – 30	
Buckling length factor about y-y		1	0.5 – 3	1.0 pinned–pinned, 0.7 fixed–pinned, 0.5 fixed–fixed, 2.0 cantilever; sway frames > 1.0.
Buckling length factor about z-z		1	0.3 – 3	Bracing at mid-height about the weak axis → 0.5.
LTB length factor		1	0.3 – 3	
Axial compression N_Ed	kN	1200	0 – …	
M_y,Ed at top	kNm	80		Strong-axis moment; same sign at both ends = single curvature.
M_y,Ed at bottom	kNm	-40		
M_z,Ed at top	kNm	5		
M_z,Ed at bottom	kNm	0		
Shear V_z,Ed	kN	30	0 – …	

Checks performed (worked example)

Check	Clause	U (example)
Cross-section: N/N_Rd + M_y/M_y,Rd + M_z/M_z,Rd ≤ 1	Eq. 6.2	0.56
Shear: V_Ed ≤ V_pl,Rd	6.2.6	0.04
Buckling about y-y: N_Ed ≤ N_b,Rd	6.3.1	0.36
Buckling about z-z: N_Ed ≤ N_b,Rd	6.3.1	0.51
LTB: M_y,Ed ≤ M_b,Rd	6.3.2.1	0.21
Eq. 6.61: N/(χ_y N_Rk/γ_M1) + k_yy M_y/(χ_LT M_y,Rk/γ_M1) + k_yz M_z/M_z,Rd ≤ 1	6.3.3(4)	0.47
Eq. 6.62: N/(χ_z N_Rk/γ_M1) + k_zy M_y/(χ_LT M_y,Rk/γ_M1) + k_zz M_z/M_z,Rd ≤ 1	6.3.3(4)	0.69

TIMBER

Timber beam

Standard: EN 1995-1-1, EN 338, EN 14080

Simply supported solid timber or glulam beam: k_mod by service class and load duration, size factor, bending with lateral-torsional buckling (k_crit), shear (k_cr), bearing (k_c,90) and instantaneous/final deflection with creep (k_def). Interactive 3D model.

Available tools: 3D model · quantities and cost · automatic design (Depth h, Width b)

Results that can be linked: γ_M, g_sw, M_Ed, V_Ed, k_h, f_m,d, σ_m,d, l_ef, σ_m,crit, λ_rel,m, k_crit, τ_d, f_v,d, k_c,90 …

Method and assumptions

- Governing combination by the largest w_d / k_mod; deflection includes shear deformation and creep (k_def).

Inputs

Input	Unit	Example	Range	Notes
Strength class		GL24h	8 …	
Width b	mm	140	38 – 400	
Depth h	mm	400	75 – 2000	
Span L	m	6	0.5 – 40	
Lateral restraint spacing (0 = span)	m	0	0 – …	Distance between restraints of the compression edge (e.g. purlins, floor joists).
Load on the compression (top) edge		yes		Destabilising: l_ef + 2h (Table 6.1).
Support length	mm	120	40 – …	
Service class		1	1 / 2 / 3	1 heated interior, 2 covered unheated, 3 exposed outdoors.
Permanent load g_k (excl. self weight)	kN/m	2	0 – …	
Variable load q_k	kN/m	3	0 – …	
Duration of the variable load		Medium-term	Permanent / Long-term / Medium-term / Short-term / Instantaneous	Imposed floor load: medium; snow: short (Cyprus); wind: short/instant.
ψ2 of the variable load		0.3	0 – 1	
Limit w_inst (variable) = L / …		300	100 – …	

Limit w_net,fin = L / …		250	100 – …	

Checks performed (worked example)

Check	Clause	U (example)
Bending: σ_m,d ≤ f_m,d	Eq. 6.11	0.57
LTB: σ_m,d ≤ k_crit f_m,d	Eq. 6.33	0.57
Shear: τ_d ≤ f_v,d	Eq. 6.13	0.4
Bearing: σ_c,90,d ≤ k_c,90 f_c,90,d	Eq. 6.3	0.38
w_inst,Q ≤ L/300	Table 7.2	0.32
w_net,fin ≤ L/250	Table 7.2	0.63